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GSFC NEON Series-1 CMO

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Near Earth Orbit Network (NEON) Series-1 Project, Code 473

Near Earth Orbit Network (NEON) Series-1 Project Sounder for Microwave-Based Applications Performance Requirements Document

NEAR EARTH ORBIT NETWORK NEON
NOAA / NASA

Goddard Space Flight Center
Greenbelt, Maryland

Near Earth Orbit Network Series-1 Project Sounder for Microwave Based Applications
Performance Requirements Document

Review/Signature/Approval Page

Prepared by:

Lisa McCormick
ATMS Instrument Systems Engineer
LEO Programs Division
NASA Goddard Space Flight Center Code 473

Approved by:

Jessica Knizhnik
SMBA Instrument Manager
LEO Programs Division
NASA Goddard Space Flight Center Code 473

Roger Clason
Deputy Program Manager
NEON Program
NASA Goddard Space Flight Center Code 470

Electronic Approval available on-line at: https://ipdtdms.gsfc.nasa.gov/frontmenu_dsp.cfm

Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

LEO/NEON Configuration Management Office
NASA/GSFC
Code 473
Greenbelt, MD 20771

Change History Log

Revision	Effective Date	Description of Changes
Revision –	February 7, 2023	This version incorporates 470-CCR-22-0372 which was approved by the SMBA ERB on December 22, 2022, and by the SMBA CCB on February 7, 2023.
Revision A	December 20, 2024	This version incorporates 470-CCR-24-0402 which was approved by the SMBA ERB on March 20, 2024, and incorporates NS1-CCR-0001 by the NS1 SMBA CCB on the effective date shown. The Document number has changed from 473-SMBA-RQMT-00004 to NS1-SMA-REQ-0001 as the document has transferred from the Management Information System (MIS) to the Technical Data Management System (TDMS). The Document name has changed from SMBA Instrument Performance Specification Document (PSD) to SMBA Instrument Performance Requirements Document (PRD). Revised the following: Requirement identifiers and numbering, Section 1.0 and its subsections, Section 4.0 and its subsections, Sections 5.1, 5.4, 5.5, 5.6, 5.7, Section 7 and its subsections & Section 8.2. Added Table 6.2-2. Moved Acronym List from Appendix A to Appendix B. Added Appendix A VCRM. Updated the following: Tables 2.1-1, Table 2.2-1, Section 3.1, Figure 3.1-1 Table 6.2-1, & Acronym List Editorial changes throughout.
Revision B	April 17, 2025	This version incorporates NS1-CCR-0009 & NS1-CCR-0011 which was approved by the NS1 SMBA CCB on the effective date shown. Figure 5.4.2-1 Pointing Terminology: FROM: Accuracy (Control) TO Accuracy

		Pointing Terminology definitions FROM Pointing Accuracy (Pointing Control) TO Pointing Accuracy Table 5.4.2-1 End-to-End Pointing Knowledge FROM +/-290 arcsec 3sigma TO +/-265 arcsec 3 sigma Table 5.4.2-1 Spacecraft Pointing Knowledge FROM +/-75arcsec 3sigma TO +/-112 arcsec 3sigma Editorial change all instances of climate to weather.

Table of Deviations/Waivers

Item No.	Deviation/Waiver	Waiver Approved	Effectivity

Table of TBDs/TBRs/TBSs

Item No.	Location	Summary	Individual/ Organization	Due Date
1	2.1	Table 2.1-1 Applicable Document List - NS1-SMBA-xxx-xxxx - NEON Series-1 Spacecraft to SMBA Instrument Interface Control Document (TBS)	Instrument Provider	
2	2.1	Table 2.1-1 Applicable Document List - NS1-SMBA-xxx-xxxx - NEON Series-1 Spacecraft to SMBA Instrument Mechanical Interface Control Document (TBS)	Instrument Provider	
3	4.1	Table 4.1-2 Note [6] Minimum resolution for RFI detection needs (TBR)	ISE	
4	4.2	SMBA-14 RFI severity levels (TBD)	ISE Instrument Provider	
5	4.3	SMBA-18 The calibration accuracy uncertainty of the channels (TBS) K.	Instrument Provider	
6	4.7	SMBA-33 The striping index 1.1 (TBR) for all heritage channels and 1.2 (TBR) for all non-heritage channels.	ISE	
7	4.7	SMBA-34 The inter-channel noise correlation coefficient 0.15 (TBR) for all channels.	ISE	
8	4.8.4	SMBA-46 The total scan period shall be TBS seconds ±TBS msec.	Instrument Provider	
9	4.9	SMBA-50 SRF magnitude (gain) knowledge (TBD)	ISE Instrument Provider	
10	4.9	SMBA-51 SRF frequency knowledge (TBD)	ISE Instrument Provider	
11	5.1	SMBA-104 Allowable heat transfer to the spacecraft during science operations is not to exceed 175 W (TBR) at an interface temperature of +15 °C (TBR).	ISE	
12	5.1	SMBA-105 Uncompensated momentum contribution of ±0.3 N-m-sec per axis (TBR).	ISE	
13	5.4	The initial data rate allocations of the instrument shall not exceed 2 Mbps orbital average and not exceed 10 Mbps peak (TBR).	ISE	

14	5.4.2	Table 5.4.2-1 Mission End-to-End Pointing Requirements: Pointing Accuracy (TBR), Pointing Knowledge (TBR)	ISE	
15	5.6.1	Table 5.6.1-5 Suggested Instrument Calibration Maneuver Requirements (TBS) and SMBA-126	Instrument Provider	
16	5.6.1.3.1	Figure 5.6.1-2 Typical Spacecraft Configuration for Earth-Pointing Safe Mode (TBR)	ISE	
17	5.6.1.4	Figure 5.6.1-4 Typical Spacecraft Configuration for Engineering Mode and Science Operations Mode (TBR)	ISE	
18	5.6.1.4	Spacecraft ECI to spacecraft Control Frame and/ or a time-tagged spacecraft ephemeris (ECEF) with a minimum frequency of 8 Hz (TBR) if required by the instrument to meet performance requirements.	ISE	
19	5.7.2	SMBA-153 Worst case expected launch load of 15 g independently in each axis (TBR).	ISE	
20	5.7.3.3.1	SMBA-171 Open loop gain margin no less than 12 dB within ±10% of structural mode frequencies, and no less than 6 dB at other frequencies. (TBR)	ISE	
21	6.2	Table 6.2-1 Radiated Susceptibility Levels Due to Factory/Transport, Launch Site, Launch Vehicle, Ascent, and On-Orbit Phases (TBR), values (TBR), Note [6] (TBR)	ISE	
22	6.2	Table 6.2-2 Radiated Susceptibility Levels Due to In-Band Sources in On-Orbit Phases (TBR/TBS) and values (TBS)	ISE Instrument Provider	
23	7.3	Table 7.3-1 Instrument and Subsystem Level Test Applicability Matrix (TBR)	ISE	
24	7.3.1	Table 7.3.1-3 Intentional and Unintentional Radiated Electric Field Emissions Limits at the SMBA Instrument Interface, 18 to 235 GHz (TBR) and values (TBR)	ISE	
25	7.3.1	Table 7.3.1-4 SMBA Receive Sensitivity (TBR), values (TBR), and Notes (TBS)	ISE Instrument Provider	
26	7.3.1.4	Receive Sensitivity (TBD) degrees from the aperture boresight(s)	ISE Instrument Provider	

27	7.3.1.4	Table 7.3.1-4 SMBA Receive Sensitivity (TBR) and values (TBR/TBD)	ISE Instrument Provider	
28	7.3.2	Table 7.3.2-1 Spacecraft Transmitter Frequency Susceptibility Test Levels (TBS/TBR) and values (TBS/TBR)	ISE Instrument Provider	
29	7.3.2.1	SMBA-232 For the Satellite RS103 Test (on-orbit configuration), the levels in Table 6.2-1 (TBD) (Columns 3 and 4)	ISE Instrument Provider	
30	7.3.2.1	SMBA-234 For the Satellite RS103 Test (Launch Configuration), the limits in Table 6.2-1 (TBD) (Columns 1 and 2)	ISE Instrument Provider	
31	7.3.3.2	Figure 7.3.3-1 Common Mode Conducted Emissions (CMCE) Limit [TBS]	Instrument Provider	
32	7.6.3	SMBA-299 outgas at a rate greater than 4.62E-7 gram/second (TBS).	Instrument Provider	
33	Appendix A	Verification Cross Reference Matrix (VCRM) [TBS]	Instrument Provider	

Table of Contents

List of Figures

List of Tables

1 INTRODUCTION

1.1 Purpose

This Sounder for Microwave-Based Applications (SMBA) Instrument Performance Requirements Document (PRD) specifies instrument-level functional, design, performance, integration, test, and delivery requirements for the instrument provider.

1.2 Document Scope

This specification establishes the requirements for the SMBA instrument. The scope of the PRD includes all functional, design, performance, integration, test, and delivery requirements for the instrument, but does not include mission-specific accommodation and interface requirements for the instrument on the spacecraft. The instrument provider is responsible for developing and maintaining a hierarchy of lower-tier specifications that derive and allocate requirements to subsystems and components.

1.3 NEON Program and NEON Series-1 Project Overview

The Near Earth Orbit Network (NEON) Program, in the National Environmental Satellite, Data, and Information Service (NESDIS) Office of Low Earth Orbit (LEO) Observations provides a framework to provide global measurements of Earth systems. These environmental observations will be able to support a wide variety of atmospheric, terrestrial, marine, and polar applications. The environmental observation data is required by the numerical weather prediction (NWP) models, fire and flood models, atmospheric chemistry observations, and multiple land imagery products that have been a crucial piece of the National Oceanic and Atmospheric Administration (NOAA) strategic goal to create a "Weather Ready Nation" and which provide essential information to NOAA's broader environmental stewardship mission. Following the recommendations from the 2017 NOAA Satellite Observation System Architecture (NSOSA) study, the NEON Program intends to use a combination of small and medium satellite platforms in a disaggregated architecture. This approach will provide an efficient means of filling gaps quickly and taking advantage of emerging remote sensing technologies.

To execute a disaggregated LEO architecture and other NSOSA recommendations, the NEON Program plans to:

- Use industry’s significant investment of funding, expertise, and innovation in space and space systems technology.
- Deploy NOAA observation system assets where and when they are most needed, enabled by shorter development timelines and more frequent launches.
- Leverage smaller instruments, satellites, and evolving small launch vehicles; and
- Achieve greater agility to incorporate continuous advancement, using new business models and partners.

The NEON Program will accomplish these goals by continuing to leverage the multi-decadal partnership between NOAA and the National Aeronautics and Space Administration (NASA).

The NEON Series-1 Project is an initial operational project for the NEON Program featuring the development, integration, test, launch, and science operations of the SMBA instrument – a successor to the Advanced Technology Microwave Sounder (ATMS) instrument, versions of which are on each of the Joint Polar Satellite System (JPSS) satellites. SMBA will allow the NEON Program to provide continuity with the existing JPSS-era microwave sounder measurements to enable advances through the incorporation of hyperspectral microwave sounding data that is crucial to numerical weather prediction models.

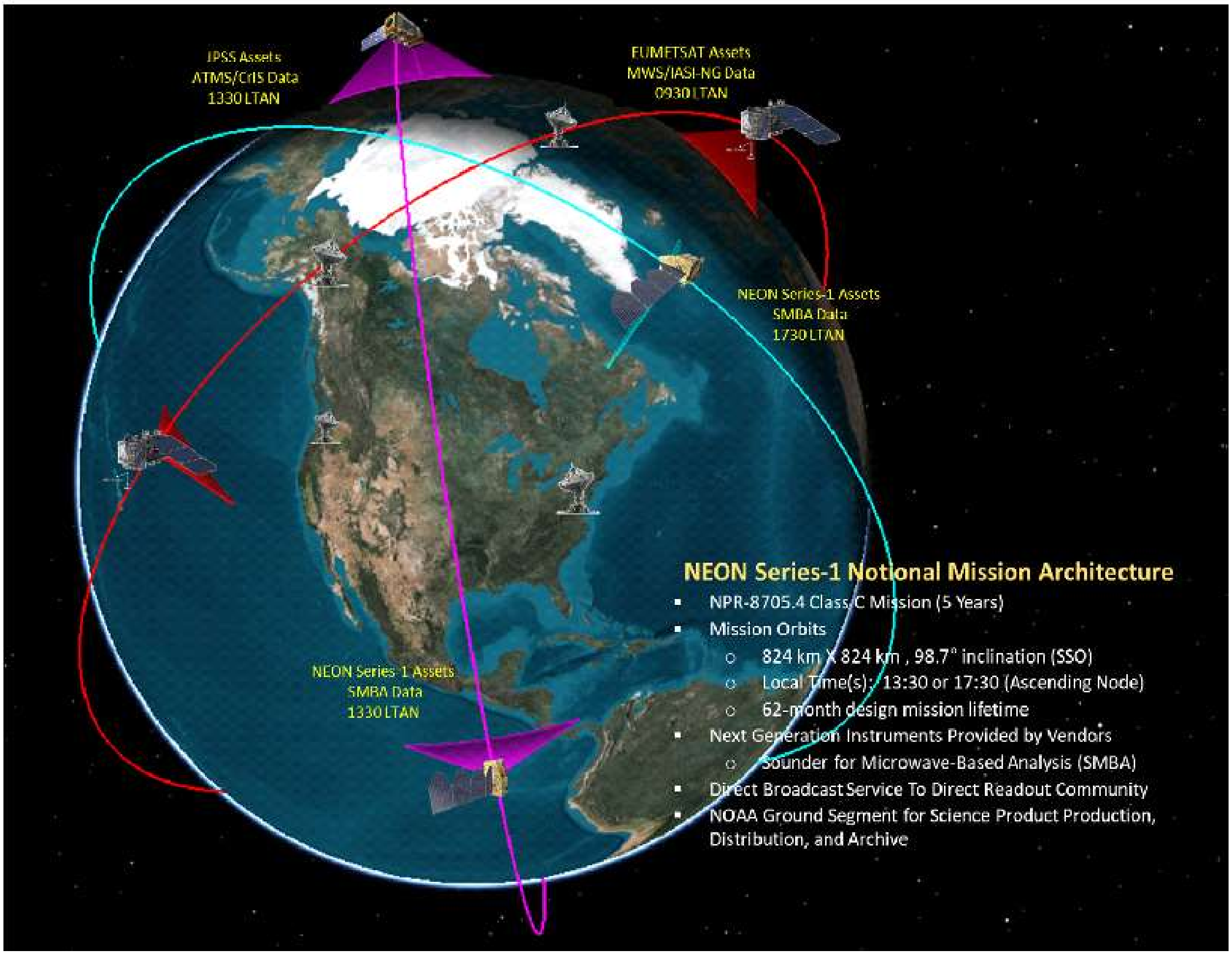

Figure 1.3-1. Notional NEON Series-1 Project Architecture

Utilizing elements and best practices of the commercial space industry to meet federal program level requirements, the NEON Series-1 Project will demonstrate the NEON Program can provide operational hyperspectral microwave sounding observation capabilities with a small satellite that can be flexibly deployed to a variety of operational orbits. This first operational project, consisting of four individual flight missions to two distinct sun synchronous orbits (local time of ascending nodes of 1330 and 1730), will demonstrate the NEON disaggregated mission portfolio that provides the program with the flexibility of placing hyperspectral microwave sounders in either operational orbit (1330 or 1730 Local Time Ascending Node (LTAN)), at any desired launch cadence as the program needs to support hyperspectral microwave sounder needs.

NEON Series-1 Project will also leverage mission assurance practices relative to established commercial New Space missions and establish a referenceable knowledge base of rapid procurement practices that will benefit the future environmental observation missions planned in the high launch tempo NEON Program.

The NEON Series-1 Project is Class C as per NASA Procedural Requirement (NPR) 8705.4, which applies the appropriate NASA management controls, systems engineering processes, mission assurance requirements and risk management processes. Select areas of redundancy should be considered that offer reasonable improvement to mission success.

1.4 The Sounder for Microwave-Based Applications

The Sounder for Microwave-Based Applications (SMBA) instrument is the principal instrument aboard the NEON Series-1 satellite. The instrument (hereafter referred to as the 'SMBA instrument' or simply the 'instrument'), is a cross-track microwave sounder that collects hyperspectral microwave radiometric data that are used to calculate the vertical distribution of temperature (T), moisture (q), and pressure in the Earth's atmosphere.

The instrument measures atmospheric radiances within the microwave range of approximately 20-300 GHz over a swath width of approximately 2,500 km from an altitude of 824 km.

Hyperspectral capability indicates fine spectral resolution, which is beneficial for both science and radio frequency interference (RFI) detection capability. Hyperspectral microwave capability refers to microwave sounding systems with approximately 100 or more channels. The increased number of channels results in a finer spectral resolution than is achieved by traditional microwave sounding techniques over the same bandwidth.

Instrument data is acquired continuously, stored on-board the satellite, and subsequently downlinked to a ground network for capture, preprocessing, and routing to meteorological centers.

The instrument produces Raw Data Records (RDR) in an uncalibrated form. Engineering, instrument telemetry, science, radiometric, and spectral calibration data are provided to enable the RDRs to be calibrated and geolocated. Additionally, diagnostic RDR data may be acquired. The data are sent to the NOAA Ground System where they are used as input to science algorithms that generate Temperature Data Records (TDR), Sensor Data Records (SDR), and Environmental Data Records (EDR).

The instrument system performance requirements are provided in subsequent sections.

1.5 Requirements Definitions

Throughout this PRD:

a. Contractually imposed mandatory requirements are identified by “shall” statements with requirement identification numbers of the form SMBA-XXXX. Any deviation from these requirements requires approval of the Government Contracting Officer.

b. Preferred good practices are identified by "should" statements. "Should" designates performance, which is requested, but not mandatory. Noncompliance of a "should" statement does not require Government approval but shall be addressed at the earliest applicable project milestone.
c. Permission is identified by "may" or "can" statements.
d. Expected outcome or action is identified by "will" statements.
e. The term "instrument" refers to the SMBA instrument which may be constructed of one or more "components" that are individually mounted to the spacecraft. If the instrument is a single assembly, then the term "component" is synonymous with "instrument." If not, then the term "component" is applicable to each severable unit of the instrument assembly.
f. The term "spacecraft bus" refers to the NEON Series-1 spacecraft without instruments.
g. The term "satellite" refers to the NEON Series-1 spacecraft bus integrated with the flight instrument(s). The term "satellite" is equivalent to the term "observatory" used in other NASA and NEON Program documents.
h. The term "spacecraft" may refer to the spacecraft bus.
i. The term "instrument provider" refers to the SMBA instrument contractor with whom the government has entered into a contract to execute the SMBA instrument Statement of Work (which is titled "Sounder for Microwave-Based Applications (SMBA) Instrument Statement of Work (SOW) (NS1-SMBA-SOW-0002)") and satisfy the requirements of this PRD.
j. The term "government" represents the appropriate NOAA and/or NASA NEON Program or NEON Series-1 Project management office or, where "the government" is indicated as providing a function for or interface to the SMBA instrument provider, the function or interface may be provided by government support services contractors.
k. The term "TBR" (to be reviewed) means that the stated information will be reviewed for appropriateness jointly by the instrument provider and the government and the information may be changed prior to final definition by the government during the course of the contract.
l. The term "TBD" (to be determined) indicates further research or analysis is needed to determine the information during the course of the contract.
m. The term "TBS" (to be supplied) indicates information that will be supplied/specified during course of instrument's development.

2 APPLICABLE/REFERENCE DOCUMENTS

2.1 Applicable Documents

The applicable documents listed below are called out within individual requirements specified in the PRD. All or portions of these documents apply directly to the performance required and contain provisions that constitute requirements of this PRD. In the event of a conflict between the language in an applicable document and in this PRD, the PRD will govern.

Table 2.1-1. Applicable Document List

Document Number	Revision	Document Title
NS1-SMBA-REQ-0002	A	NEON SMBA Instrument Mission Assurance Requirements
NS1-SMBA-xxx-xxxx	*	NEON Series-1 Spacecraft to SMBA Instrument Interface Control Document
NS1-SMBA-xxx-xxxx	*	NEON Series-1 Spacecraft to SMBA Instrument Mechanical Interface Control Document
470-LEO-ANYS-00018	Rev -	Radiation Environment for the NEON Program
ISO 14644-1	May 1, 1999	Cleanrooms and Associated Control Environments
MIL-STD-461C	C August 4, 1986	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference (EMI), Revision C (August 4, 1986)
MIL-STD-461G	G December 11, 2015	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, Revision G (December 11, 2015)
MIL-STD-462	Rev - Notice 2 from May 1, 1970	Measurement of Electromagnetic Interference Characteristics (Notice 2)
NASA-STD-5001	B w/ change 3	Structural Design and Test Factors of Safety for Spaceflight Hardware
GSFC-STD-7000 Section 2.5.2.1.1 Section 2.5.2.1.2 Section 2.5.2.2.3	B	General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects
NASA-STD-8719.14	C	NASA Process for Limiting Orbital Debris
NPR 8715.6	B	NASA Procedural Requirements for Limiting Orbital Debris

*See Contract Attachment N, Applicable Documents List for current revision

2.2 Reference Documents

The reference documents listed below are referred to within the PRD and contain information relating to the work required. The documents do not contain provisions that constitute requirements of this PRD, but rather serve to amplify or clarify the requirements.

Table 2.2-1. Reference Documents List

Document Number	Revision	Document Title
ASTM E595	5/14/2021	Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment
CCSDS 301.0-B-4	November 2010	Consultative Committee for Space Data Systems (CCSDS) Time Code Formats Standard
GSFC-STD-7000	B	General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects
IEST-STD-CC1246	E	Product Cleanliness Levels and Contamination Control Program
472-00228	B	JPSS Flight Project Contamination Control Plan
JSC-CR-06-070	August 2006	Space Vehicle RF Environments
NASA-STD-5017	B	Mechanism Design Requirements
NPR 8705.4	4/29/2021	Risk Classification for NASA Payloads
NASA-STD-8719.24	Initial with Change 1, Annex with Change 2 Jun 2012	NASA ELV Payload Safety Requirements
http://outgassing.nasa.gov		Outgassing Data for Selecting Spacecraft Materials

3 GENERAL REQUIREMENTS

3.1 On-Orbit Coordinate System

The NEON Series-1 satellite's on-orbit coordinate system uses a right-hand, orthogonal, body-fixed XYZ coordinate system as follows (see Figure 3.1-1): the +Z-axis is downward towards geodetic nadir, the Y-axis is along the orbit normal plane (+Y is opposite the orbital angular momentum), and the X-axis is along the spacecraft velocity vector (+X toward the direction of spacecraft travel).

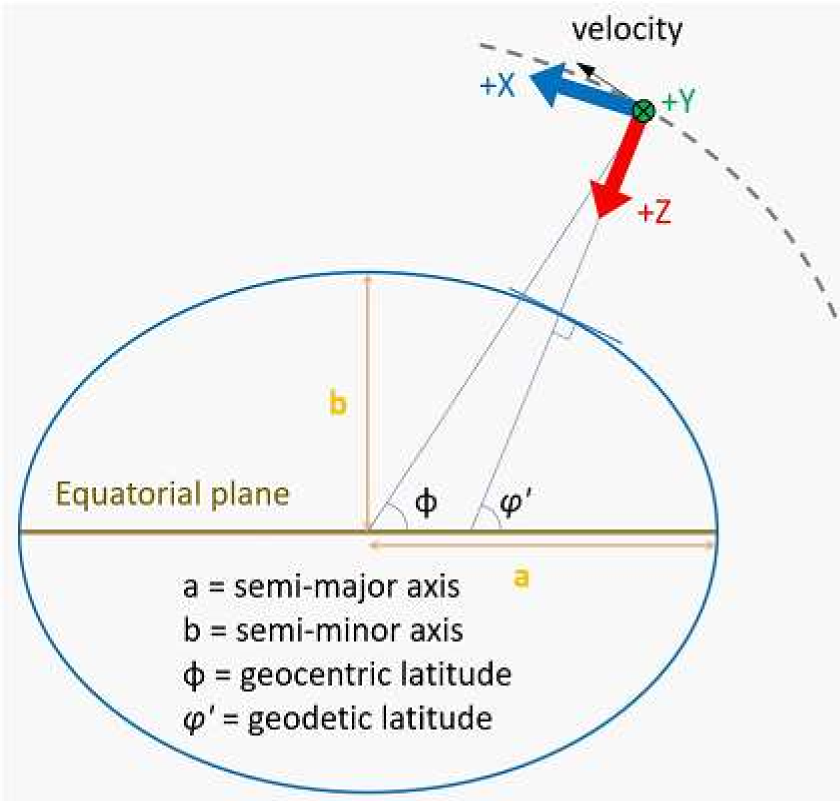

Figure 3.1-1. On-Orbit Coordinate Systems

NEON Series-1 Project uses J2000-referenced orbital elements for attitude knowledge and the WGS84 geodetic reference system for environmental data geolocation.

The instrument should use the international system of units, Système Internationale (SI), unless precluded by design or manufacturing heritage. All interface documents should provide units in SI and English. The primary (as-designed) units should be displayed above the converted units with the latter in parentheses.

3.2 Mission Time Convention

SMBA-1	The instrument shall time stamp packets in the secondary header using Coordinated Universal Time (UTC), including the leap-second convention, also known as CCSDS Day Segmented Time Code (CCSDS 301.0-B-4), for all instrument telemetry while the instrument processor is running.

Note: Telemetry frames that are generated by hardware during ascent or processor down time may format the time stamp using CCSDS Unsegmented Time Code (CCSDS 301.0-B-4).

3.3 Risk Classification and Mission Assurance Requirements

The instrument risk classification is Class C, as per NPR 8705.4, Risk Classification for NASA Payloads.

SMBA-2	The instrument provider shall comply with SMBA Instrument Mission Assurance Requirements (NS1-SMBA-REQ-0002).

Safety design requirements and requirements for the control of catastrophic and critical failures are contained in SMBA Instrument Mission Assurance Requirements (NS1-SMBA-REQ-0002).

The instrument and its ground support equipment used at the launch site will comply with the NASA-STD-8719.24 NASA Expendable Launch Vehicle (ELV) Payload Safety Requirements or other applicable range safety documentation contained in SMBA Instrument Mission Assurance Requirements (NS1-SMBA-REQ-0002).

4 SMBA SCIENCE REQUIREMENTS

In Section 4, all requirements apply over the life of the instrument.

4.1 SMBA Channel Requirements

A frequency band is a portion of the electromagnetic spectrum associated with a letter designation. For typical use in SMBA, refer to the frequencies and associated letter designation in Table 4.1-1.

Table 4.1-1. SMBA Letter Designations for Frequency Bands

Letter Designation	Frequency Range	Band Purpose
K band	20 - 25 GHz	Column Integrated Observations & Quality Control
Ka band	31 - 33 GHz	Window
V band	50 - 62 GHz	Temperature (T)
W band	85 - 92 GHz	Window
F band	108 - 123 GHz	Temperature (T)
D band	164 - 167 GHz	Window
G band	173 - 193 GHz	Humidity (q)
J band	200 - 235 GHz	Window & Ice Cloud Detection

The SMBA instrument design may cover a subset of the frequency range associated with a letter designation. The frequency range is not prescribing precise start and stop frequencies or bandwidths. The instrument frequency plan should take into consideration science benefits, RFI detection methods, downlink throughput, hardware constraints, etc.

A channel is the portion of a frequency band defined by a spectral response function (SRF). For the SMBA instrument, SRF is defined as a magnitude of energy versus frequency transfer function, where the input is at the antenna aperture and the output is at the instrument output. The output of the instrument for a channel corresponds to the integrated energy under the SRF. A channel is only within one frequency band and does not span across multiple frequency bands.

Center frequency is defined as the median frequency between the half-power points of a passband.

Bandwidth is defined as the spectral width of a passband, where the width is the frequency range between the half power points that define the start and stop frequencies. The total bandwidth of a channel is the sum of the bandwidths of the passbands comprising the channel.

Center frequency stability is defined as the maximum deviation from the center frequency over the operational life of the instrument.

Hyperspectral capability indicates a fine spectral resolution of the electromagnetic spectrum within a given specified frequency band. Fine spectral resolution enabled by a hyperspectral capability will be beneficial to address science needs and RFI detection needs (Section 4.2). While hyperspectral capabilities will allow finer spectral resolution, the instrument need not downlink all of the collected bandwidth. The downlinked channels should be a balance between data volumes and spectral resolution.

SMBA-3	Hyperspectral capability shall be provided for the frequency bands identified in Table 4.1-2.

Table 4.1-2. Hyperspectral and RFI Detection Capabilities by Band

Frequency Bands	Hyperspectral Capability	RFI Detection Capability	Hyperspectral Resolution (MHz)
K	Required[1]	Required[5]	≤ 10[6]
Ka	Required[1]	Required[5]	≤ 10[6]
V	Required[2]	Required[5]	≤1 between 56-58 GHz ≤ 10 outside of 56-58 GHz[6]
W	Required[1]	Required[5]	≤ 20[6]
F	Required[2]	Not Required	≤ 20
D	Not Required	Not Required	N/A
G	Required[3]	Not Required	≤ 100
J	Not Required	Not Required	N/A

Notes:
[1] Capability required for Window Channels continuously in operational mode.
[2] 20 or more channels required in V band and 10 or more channels in F Band for Temperature Sounding for continuously in operational mode.
[3] 10 or more channels required in G Band for Moisture Sounding for continuously in operational mode.
[4] Band Frequency Start and Stop to reflect the specifics of the instrument chosen.
[5] RFI detection capability required continuously in operational mode.
[6] Minimum resolution for RFI detection needs (TBR)

SMBA-4 For science continuity, SMBA instrument shall provide the heritage channels at the center frequencies and bandwidths identified in Table 4.1-3 with the channel performance of Table 4.1-4.

SMBA-5 In addition to the heritage channels, the instrument shall provide new hyperspectral and/or RFI detection channels within the same band that meet the channel performance of Table 4.1-4.
Note: New channels may use or reuse any frequency spectrum within same band. New channels may be formed by aggregating the minimum hyperspectral resolutions. Aggregated channel bandwidths do not need to be uniform across the frequency band.

Table 4.1-3. NEON Program Microwave Sounder Heritage Channel Requirements

Center Frequency[1] (GHz)	Bandwidth (GHz)
23.8	0.27
31.4	0.18
50.3	0.18
51.76	0.4
52.8	0.4
53.71080 54.48080	0.17 0.17
54.4	0.4

54.94	0.4
55.5	0.33
57.290344	0.33
57.073340 57.507340	0.078 0.078
56.920144 57.016144 57.564544 57.660544	0.036 0.036 0.036 0.036
56.946144 56.990144 57.590544 57.634544	0.016 0.016 0.016 0.016
56.958144 56.978144 57.602544 57.622544	0.008 0.008 0.008 0.008
56.963644 56.972644 57.608044 57.617044	0.003 0.003 0.003 0.003
88.2	2.0
114.50	1.0
115.95	0.8
116.65	0.6
117.25	0.6
117.80	0.5
118.24	0.38
118.58	0.30
165.5	3.0
176.31 190.31	2.0 2.0
178.81 187.81	2.0 2.0
180.31 186.31	1.0 1.0
181.51 185.11	1.0 1.0
182.31 184.31	0.5 0.5
229	2.0

Notes:

[1] In the Center Frequency column, each boxed row is one channel and contains one or more passbands. The passbands in each row are the passbands to be aggregated to form the heritage channel. For example, Table 4.1-3 lists one channel with two passbands centered at 176.31 GHz and 190.31 GHz. Fewer passbands in a channel may be used, provided the Temperature Sensitivity requirements are met for the heritage channel bandwidth.

SMBA-6 The center frequency stability of the channels shall be less than or equal to the values listed in Table 4.1-4.

Note: Calibration accuracy and NEDT requirements are found in Sections 4.3 and 4.4, respectively.

Table 4.1-4. NEON Program Microwave Sounder Channel Performance Requirements

Frequency Bands[1]	Center Frequency Stability (MHz)	Calibration Accuracy (K)	NEDT[2],[3] (K*sqrt(msec*MHz))
K	5	0.5	16.7
Ka	5	0.5	15.9
V	0.3	0.5	18.7
W	18	0.5	37.9
F	1	0.5	31.0
D	22	0.4	61.0
G	9	0.4	67.0
J	22	0.5	68.3

Notes:
[1] Provider may utilize any portion(s) of the band and optimize the configuration to enhance hyperspectral capability and RFI detection capability in Section 4.2.
[2] NEDT values assume:
(a) 300K scene temperature observation
(b) an integration time of 1 msec and a bandwidth of 1 MHz
(c) No radio frequency interference in measurement
(d) To scale NEDT for the actual integration time and bandwidth, divide the value by sqrt(actual integration time in msec*actual bandwidth in MHz).
[3] NEDT compliance at raw footprint level is preferred. Instrument provider to show both NEDT computed values from raw footprints and NEDT computed values after resampling or averaging.

SMBA-7 In frequency bands requiring hyperspectral capability, the hyperspectral resolution shall be as specified in Table 4.1-2.
Note: Heritage channels (Table 4.1-3) have bandwidths greater than the minimum hyperspectral resolution (Table 4.1-2) and will be formed by aggregating the minimum hyperspectral resolutions. Aggregated channel bandwidths do not need to be uniform across the frequency band.

SMBA-8 The configuration of heritage, new hyperspectral, and RFI detection channels identified in SMBA-5 shall be reprogrammable on orbit.

4.2 Radio Frequency Interference Detection Capability

The global expansion of other International Telecommunication Union (ITU) services into bands adjacent to and overlapping with SMBA bands poses a threat to the quality of the radiometric data and, therefore to the accuracy of NWP forecasts and other products. Accurate RFI detection (flagging) will allow NWP models and science community users to identify contaminated observations and act accordingly.

Table 4.1-2 requires on-board RFI Detection Capability for frequency bands that might experience RFI. Given the importance of these bands for science performance and continuity, the hyperspectral capability enhances the RFI Detection Capability of the SMBA instrument. The on-board RFI Detection Capability should incorporate flexible, repeatable, and effective methods to identify and characterize the continually changing man-made sources of interference over the SMBA mission life.

SMBA-9	Detection of Radio Frequency Interference (RFI) by man-made signals that impact operational data products, shall be reported for all spectral bands identified in Table 4.1-2. Note: man-made signals may be continuous, pulsed, narrowband, broadband, etc.

RFI detection flag should occur at the appropriate processing level to allow assessment of the RFI detection algorithm and to maximize the preservation of science data.

SMBA-10	The instrument shall be capable of on-board RFI detection simultaneously with science data collection.
SMBA-11	The instrument shall be capable of executing multiple RFI detection algorithms simultaneously.
SMBA-12	To assess RFI detection algorithms, the instrument shall be capable of collecting, storing, and relaying to ground raw data used for RFI detection. Note: Raw data refers to internal instrument data along the processing chain prior to the formation of channels. Raw data may include data in time domain and/or frequency domain.
SMBA-13	SMBA RFI detection shall have the capability to enable or disable detection functions through on-board scripts and ground commanding. Note: Examples of RFI detection functions include selecting frequency bands, selecting geolocations, and selecting type of raw data to downlink.
SMBA-14	SMBA RFI detection flag packet shall identify the algorithm, frequency range corrupted, channel(s) affected, and RFI severity level. Note: spacecraft provides time stamp and ephemeris data. RFI severity levels are TBD.
SMBA-15	SMBA instrument shall have the capability to reprogram on-board RFI detection algorithms through ground command.
SMBA-16	SMBA RFI detection capability shall report metrics on RFI detection algorithm performance (i.e., detection sensitivity, false alarms, flagging, diagnostic data, raw data, etc.).

4.3 Calibration Accuracy

Calibration accuracy is defined as the difference (error) between the brightness temperature inferred from the microwave radiometer (referred to the antenna collecting aperture) and the actual brightness temperature of an external calibration reference.

The calibration accuracy is characterized via two parameters: magnitude and uncertainty. The calibration accuracy magnitude describes the size and direction of the difference between the inferred brightness temperature and the actual brightness temperature.

The calibration accuracy uncertainty describes the tolerances associated with the magnitude value. Calibration accuracy magnitude and uncertainty contributions may be acquired on the ground and/or on orbit.

Calibration accuracy contributions include, but are not limited to, brightness temperature of calibration references and the radiometric calibration transfer function (counts to kelvins). Figure 4.3-1 shows notional calibration accuracy contributions as a reference to be considered for SMBA.

Contributions can include, but are not limited to, uncertainties due to (1) errors contained in the ground calibration tests and the test equipment, (2) the transition from the ground cold calibration reference to the on-orbit cold calibration reference, and (3) emissions from the spacecraft intercepted by the near fields of the radiometer antenna(s).

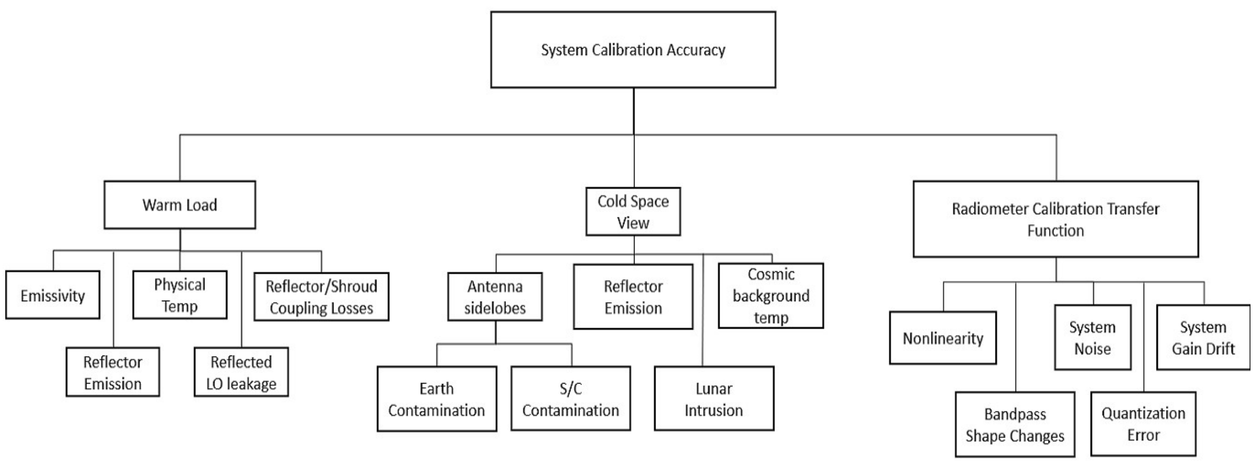

Figure 4.3-1. Notional Calibration Accuracy Contributions

Notes:

[1] A calibration reference is defined as an element that provides a signal that corresponds to a known brightness temperature. It can be a physical thing (like a blackbody; ex: on-board target or cold space). A noise diode is also an example of a reference.

[2] Traditional radiometers have used two-point calibration methodology: cold space for the cold reference temperature and an on-board blackbody for the warm reference temperature. The instrument design is not constrained to this methodology or these calibration reference types. Other types of calibration references exist such as noise diodes and cold Field Effect Transistors (FET).

[3] Proposed references that are also standards (as defined by the National Institute of Standards and Technology (NIST)) must include a description of the traceability plan that enables them to be a standard, along with an estimate of the (absolute) accuracy. SMBA follows the NIST definitions of "standard," "reference," "intercalibration," and "intercomparison."

Values for contributions in Figure 4.3-1 are characterized on the ground, on-orbit, and by analysis. Some contributions shown in Figure 4.3-1 have specific requirements associated with them because they may be a relatively large error source, can be best characterized on the ground, impact numerous samples within a scan, and/or are corrected for in the operational algorithm.

A specific calibration accuracy contribution may have dynamic and/or static characteristics. Dynamic characteristics are those that vary with time. Static characteristics are those that do not vary with time.

This results in a system-level calibration accuracy consisting of dynamic and static contributions. The time scale of one orbit captures orbital variations. Calibration accuracy is the average error over a time scale longer than one orbit.

Corrections to the data are based on the instrument design are becoming increasingly important as demands for accuracy increase. The instrument provider should have a sufficient understanding of how to make the corrections for the calibration accuracy magnitude and determine the residual uncertainty (error). Static contributions may be corrected for as a bias correction in the algorithm.

SMBA-17	The calibration accuracy magnitude of the channels shall be less than or equal to the values listed in Table 4.1-4.

SMBA-18	The calibration accuracy uncertainty of the channels shall be less than or equal TBS K.

SMBA-19	Calibration accuracy predictions shall be computed as the sum of static (bias) errors and the root-sum-square of the standard deviations of dynamic (time-variable) errors, for worst case environmental conditions.

SMBA-20	All calibration accuracy computations shall be for 1-sigma statistics properties.

SMBA-21	The brightness temperature uncertainty of the warm calibration target shall be less than or equal to +/-0.20 K.

SMBA-22	The emissivity of any on-board blackbody calibration references shall be greater than or equal to 0.99990 within an uncertainty less than or equal to +/-0.00003.

SMBA-23	The physical temperature gradient of any warm load calibration reference shall be less than or equal to 0.2 K.

SMBA-24 The physical temperature uncertainty of any on-board blackbody calibration references shall be less than or equal to +/- 0.1 K.

SMBA-25 The brightness temperature uncertainty of the cold space calibration reference shall be less than or equal to +/- 0.5 K.

SMBA-26 The near-field antenna characteristics shall be determined to sufficient accuracy such that brightness temperature contributions from spacecraft and instrument radiation have an uncertainty less than or equal to the values specified in Table 4.3-1.

SMBA-27 The far-field antenna characteristics shall be determined to sufficient accuracy to ensure that the brightness temperature contributions from Earth radiation have an uncertainty less than or equal to the values specified in Table 4.3-1.

Table 4.3-1. Antenna Brightness Temperature Uncertainty Maximum

Frequency Band	Near-field[1],[2] (K)	Far-field[1],[3] (K)
K	+/- 0.075 K	+/- 0.35 K
Ka	+/- 0.075 K	+/- 0.35 K
V	+/- 0.05 K	+/- 0.20 K
W	+/- 0.05 K	+/- 0.20 K
G	+/- 0.05 K	+/- 0.20 K
J	+/- 0.05 K	+/- 0.20 K

Notes:
[1] Distance, d, is the aperture size of the antenna and λ is the wavelength measured by the antenna.
[2] Distances less than $2d^2/\lambda$ from the antenna are defined as near-field.
[3] Distances greater than or equal than $2d^2/\lambda$ from the antenna are defined as far-field.

4.4 Temperature Sensitivity

Brightness temperature sensitivity (noise equivalent delta temperature (NEDT)) is defined as the standard deviation of the radiometer output brightness temperature in kelvins (K) incident at the antenna collecting aperture when the antenna is viewing a 300 K uniform and stable target.

Integration time is defined as the time interval over which the radiometric signal is collected for a given channel.

SMBA-28 The temperature sensitivity of the channels shall be less than or equal to the values listed in Table 4.1-4.

SMBA-29 The brightness temperature measurement of an Earth scene or calibration reference shall be independent of any previous measurements.

4.5 Channel Dynamic Range

SMBA-30	The dynamic range of the instrument shall envelope the brightness temperature range of Earth scenes, calibration references, and be sufficient to meet the RFI requirements in Section 4.2.

4.6 Channel Linearity

Nonlinearity is the departure from the expected response of an ideal linear radiometer. Channel nonlinearity is determined from a second-order or higher fit to the calibration accuracy data obtained during thermal-vacuum calibration testing. Nonlinearity is one-half the maximum deviation from the linear response due to the polynomial curve from the low end of the dynamic range to the high end of the dynamic range, with the polynomial curve and linear response intersecting at the low end and high end. Residual nonlinearity is the residual error after applying a nonlinearity correction to the individual calibration accuracy points.

SMBA-31	As input to the computation of residual nonlinearity, the nonlinearity shall be computed for each channel from accuracy measurements made during thermal vacuum ground calibration testing.

SMBA-32	The residual nonlinearity, after calibration corrections to any measurement within the dynamic range, shall be less than or equal to 0.1 K for all channels.

4.7 Noise Characteristics

Differences in the along-scan vs. along-track noise characteristics can lead to undesirable linear artifacts in images known as striping. The striping index can be used to characterize the magnitude of striping. The striping index is defined as the ratio of the along-track variance in brightness temperature divided by the cross-track variance in brightness temperature.

SMBA-33	The striping index shall be less than or equal to 1.1 (TBR) for all heritage channels and 1.2 (TBR) for all non-heritage channels.

Inter-channel noise correlation is measured by correlating the noise signal in one channel to the concurrent noise signal in another channel while viewing the same Earth scene or calibration reference, resulting in a correlation coefficient.

SMBA-34	The inter-channel noise correlation coefficient shall be less than or equal to 0.15 (TBR) for all channels. Note: A channel correlates perfectly with itself and has an inter-channel noise correlation coefficient of 1.00.

4.8 Beam Characteristics

Each frequency band of the instrument is considered to form an antenna beam. In the following sections, if only one beam is discussed it refers to any and all beams.

The antenna beam electrical boresight is defined as the direction of maximum gain of the antenna.

The term "Beam Position" means the position of a "Beam-Center," which is defined as the position of the antenna beam electrical boresight at the mid-point of the integration time.

Beamwidth is defined as the angular width between the half-power points relative to the maximum gain of the antenna in the electric field plane that contains the boresight vector.

Antenna main beam is defined as the angular region around the boresight within 2.5 times the beamwidth of the antenna. (Note: Not +/-2.5 times.)

Antenna main beam efficiency is defined as the ratio of all energy received within the main beam, i.e., including cross-polarized energy, to the total of all energy received by the antenna.

SMBA-35	The beamwidth in any plane containing the electrical boresight axis shall be within plus or minus 10% of the nominal beam width.

SMBA-36	Antenna main beam efficiency shall be no less than 95.0% at all frequency bands and all beam positions.

4.8.1 Beam Pointing

From an atmospheric sounder, NWP modelers and other users need a series of measurements over a range of altitudes to construct an atmospheric profile of temperature or moisture (Atmosphere Vertical Temperature Profile (AVTP), Atmosphere Vertical Moisture Profile (AVMP)) where a profile is a snapshot of the thermodynamic or moisture state along a line from the top to the bottom of the atmosphere.

There are important physical connections between the state at one altitude and the state just above and below that altitude; likewise, the measurement at one altitude overlaps with the measurement just above and below that altitude. The states are connected along a line from the top to the bottom of the atmosphere. Therefore, a sounder will make measurements within the projected beam along a line from the sensor to the Earth’s surface. Different frequency bands and channels yield measurements at different altitudes.

In order for those measurements at different altitudes to all fall long the same line, the beams for all bands will point along the same line of observation from the sensor to the Earth’s surface (spatial simultaneity) and at the same time since scanning at orbital velocity will move to a new observation line (a new profile) within milliseconds (temporal simultaneity).

Note that the spatial simultaneity requirement applies along the entire line from the top to the bottom of the atmosphere. Just requiring the footprints of all the beams to be coincident only constrains the beams at the Earth’s surface. Both endpoints of the line (beams) should be as coincident as possible.

SMBA-37	For any given beam position, all antenna beam electrical boresight axes of the instrument shall point in the same direction and intersect the same location on the Earth's surface, subject to the pointing accuracy requirements, within 1/10 of a footprint integration time.

4.8.2 Channel Rejection

SMBA-38	Each channel shall have a gain that is no greater than 40 dB below the band-center value for all frequencies outside of a band ±0.70 times the specified half-power bandwidths.

All channels that use upper and lower mixer sideband signals should employ stopbands to reduce local oscillator noise to levels compatible with meeting the sensitivity and the calibration accuracy requirements.

4.8.3 Gain

Passband gain is defined as the overall gain of the instrument from the antenna aperture to the instrument output, averaged over the passband bandwidth.

SMBA-39	The gains of all passbands within any one channel shall be within ±1 dB of each other.

4.8.4 Scan Characteristics

SMBA-40	The instrument shall provide contiguous or overlapping spatial sampling (3dB contours of the antenna patterns) both along-track and cross-track when operated in the mission orbit. Note: Nyquist spatial sampling is an acceptable option, as well.
SMBA-41	The instrument shall provide global coverage greater than or equal to 95% when operated in the mission orbit.
SMBA-42	The instrument shall have a minimum spatial horizontal resolution at nadir of 32 km (T) and 16 km (q) when operated in the mission orbit.
SMBA-43	The instrument shall have a minimum vertical reporting interval of 4 km (T and q).
SMBA-44	The instrument shall integrate while scanning (i.e., not step scan).
SMBA-45	The instrument antenna beam electrical boresight shall scan cross-track to the satellite motion.
SMBA-46	The total scan period shall be TBS seconds ±TBS msec.

SMBA-47	Instrument shall synchronize its scan start time with a timing signal it receives from the spacecraft command and telemetry (C&T) bus.

SMBA-48	The time offset from the rising edge of the spacecraft timing signal to the start of beam position number one of the instrument scan shall have an accuracy of ±1 msec.

SMBA-49	The offset shall be settable over a range of zero to the scan period.

4.9 Spectral Response Function

The Spectral Response Functions (SRF) are created via a combination of measurement and analysis. For a channel having multiple passbands, the SRF of each passband is unique and must be independently ascertained.

Measuring the SRF creates a one-to-one mapping between the end-to-end gain values (from the antenna aperture through the instrument output) and the RF frequencies of radiation relevant to each channel.

SRFs are used to generate: (1) Band absorption coefficients for radiative transfer models used by environmental retrievals algorithms and NWP models. NWP models continue to require increasingly accurate SRF measurements. (2) Band conversion coefficients in the SDR algorithm that calculate warm/cold target radiance from their physical temperatures and Earth spectral brightness temperature from Earth radiance.

SMBA-50	The magnitude (gain) knowledge of a SRF shall be within +/- TBD dB.

SMBA-51	The frequency knowledge of a SRF shall be within +/- TBD Hz

4.10 Calibration

SMBA-52	Instrument shall perform in-flight calibration for all channels.

SMBA-53	The instrument shall be designed such that the ground calibration needs to be performed only once for each delivered instrument.

SMBA-54	No ground calibration shall be required after prolonged storage before launch.

SMBA-55	All channels shall use an end-to-end calibration method, including the full optical chain.

SMBA-56	The cold calibration target shall be the cosmic microwave background radiation, also referred to as cold space.

SMBA-57	During cold calibration, the instrument shall view the cold calibration reference directly without the use of additional optical elements.

Note: Requirements SMBA-58 through SMBA-99 are reserved for future instrument science requirements pertaining to specific implementation approaches.

5 NEON SERIES-1 INSTRUMENT REQUIREMENTS

The following paragraphs contain requirements extracted from the NEON Series-1 Project-level requirements documentation and are directly traceable to NOAA's Mission Requirements.

5.1 Instrument Accommodations

SMBA-100	The instrument shall comply with the accommodation requirements, in accordance with the requirements delineated herein and the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx), including launch vehicle mechanical, electrical, environmental requirements, and constraints.
SMBA-101	The instrument shall define and maintain static and dynamic clearances to the respective instrument envelope(s), define and maintain the instrument operational, calibration, and thermal field-of-views, and define integration and test target accommodations in the Spacecraft-to-SMBA Instrument MICD (NS1-SMBA-xxx-xxxx).
SMBA-102	The instrument mass, including blankets and instrument harnessing, shall not exceed 150 kg. Mass properties, including center of mass and moments of inertias, will be defined in the Spacecraft-to-SMBA MICD.
SMBA-103	The instrument shall not exceed an orbit average power load of 295 W in any instrument operational mode. The instrument power allocation per mode and peak power will be defined in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).
SMBA-104	While an adiabatic interface is preferred, the initial allowable heat transfer to the spacecraft during science operations is not to exceed 175 W (TBR) at an interface temperature of +15 °C (TBR). The specific thermal interface needs (inclusive of margin) of the instrument shall be specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).
SMBA-105	The instrument shall not exceed an uncompensated momentum contribution of ±0.3 N-m-sec per axis (TBR). The specific level of on-orbit disturbances induced to the spacecraft across the instrument-to-spacecraft mounting interface shall be specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).

5.2 Mission Orbit Requirements

SMBA-106	The instrument shall operate for the mission lifetime in a polar sun-synchronous orbit with the following characteristics: Nominal Altitude: 824 km +/- 17 km Nominal Inclination: 98.71 degrees +/- 0.10 degrees Ground Track Repeatability Accuracy: +/- 20 km at the equator Ground Track Repeat Cycle: 16 days

SMBA-107	The instrument shall operate for the mission lifetime in the polar sun-synchronous orbit defined in SMBA-106 when deployed/maintained orbit in either of the following local time of the ascending nodes (LTANs): Nominal Ascending Equator Crossing Time: 1330 (local time) +/- 10 minutes or Nominal Ascending Equator Crossing Time: 1730 (local time) +/- 10 minutes

5.3 Mission Lifetime, Commissioning, and Storage Requirements

SMBA-108	The instrument shall be designed for a mission lifetime of 5 years from completion of on-orbit commissioning.

SMBA-109	The instrument shall be capable of a 10-year ground storage period that includes routine maintenance and monitoring activities to ensure no loss of functional capabilities.

SMBA-110	The instrument shall be capable of being placed in a ground storage state that requires minimal intervention by personnel for up to 30-day periods.

5.4 Mission Data Requirements

The spacecraft command and data handling (C&DH) subsystem collects instrument health & status telemetry, instrument memory dump, and mission data packets through the command & data interface. The C&DH subsystem multiplexes these packets and attaches appropriate protocol for downlink and storage. The need for following functions will be determined jointly by the instrument and spacecraft providers and defined in the applicable instrument ICD.

a. Spacecraft-to-Instrument transfers consisting of real-time ground commands, stored commands, memory loads, state of health indications, and auxiliary data (e.g., time code).
b. Instrument-to-Spacecraft transfers consisting of, for example, mission data (including auxiliary data received from the spacecraft), instrument health and status telemetry, indication of instrument mode transitions, instrument diagnostic data, memory dumps, and Survival Mode temperatures (not part of the data bus).

The instrument average data rate is defined to be the total data transmitted for one orbit divided by the orbit period. The instrument peak data rate is defined as the maximum data volume divided by the collection period.

SMBA-111	The initial data rate allocations of the instrument shall not exceed 2 Mbps orbital average and not exceed 10 Mbps peak (TBR). The instrument average and peak data rates per instrument mode will be specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).

SMBA-112	All data that comprises the RDR shall be downlinked uncompressed or using lossless compression.

SMBA-113 Receipt of individual instrument commands shall be verifiable via instrument health and status telemetry.

5.4.1 Mission Data Geolocation Requirements

The instrument pointing accuracy is the difference between the actual and the intended antenna boresight direction. This includes the as-built static error and the effects of all structural, thermal, and environmental sources that induce changes in the boresight alignment.

SMBA-114 The known geolocation of the instrument boresight, referenced to the center of the effective field of view for any channel, shall be within 1.50 km (radial, 3 sigma) of the true location of that field of view at nadir on the WGS84 reference ellipsoid, at any time during nominal operations.

5.4.2 Instrument Pointing, Accuracy, and Stability Requirements

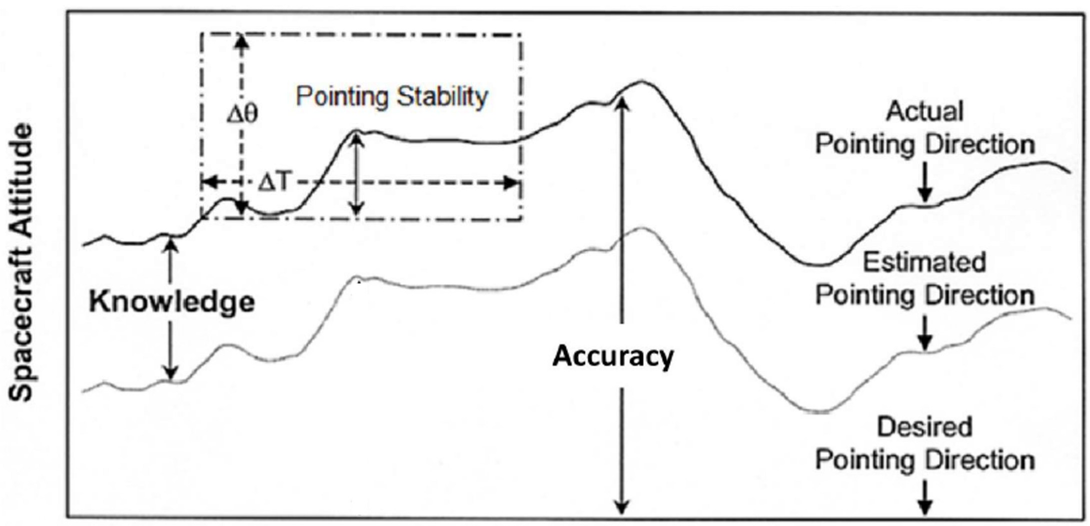

Figure 5.4.2-1. Pointing Terminology

Pointing terminology definitions are illustrated in Figure 5.4.2-1 and are defined as follows for each spacecraft axis:

- Pointing Knowledge - The angle between the actual pointing direction and the estimated pointing direction.
- Pointing Accuracy - The angle between the actual pointing direction and the desired pointing direction.
- Pointing Stability - The peak-to-peak variation in the actual pointing direction over a time interval.
- Accuracy and Knowledge terms consist of three error categories:
  i. Bias: errors that remain constant over the duration of the mission.
  ii. Drift: A long term variation of pointing direction.

iii. Jitter: A short term variation of the pointing direction.

- Static Errors - Errors that stay constant over mission time. Bias errors are static errors.
- Dynamic Errors - Errors that vary over time. Drift and Jitter errors are dynamic errors.
- Pointing Stability is due to dynamic error sources Drift and Jitter.
- 3 Sigma - A set of values is considered to meet a 3-sigma requirement if no fewer than 99.73% of the values are within the specified limits of the requirement.

All contributions to pointing uncertainties relating the spacecraft attitude measurements made in the spacecraft attitude reference (AR) frame to the alignment of the instrument interface (mounting surface datums) are allocated to the spacecraft.

All contributions to pointing uncertainties relating the instrument interface (mounting surface datums) to the instrument boresight are allocated to the instrument.

Contributions to pointing uncertainties resulting from clearances across the mounting interface (pin/fastener to hole tolerance) at the instrument interface (mounting surface datums) are allocated to the instrument.

SMBA-115 End-to-End pointing accuracy, pointing knowledge, and pointing stability shall meet the requirements specified in Table 5.4.2-1 per axis.

Table 5.4.2-1. Mission End-to-End Pointing Requirements[1]

End-to-End Pointing Requirement	Requirement (@ antenna boresight)
Pointing Accuracy	±2400 arcsec 3σ (TBR)
Pointing Knowledge	±265 arcsec 3σ (TBR)
Pointing Stability	N/A
Spacecraft Contributions	**GN&C Requirement (@ AR)**
Pointing Accuracy	± 900 arcsec 3σ
Pointing Knowledge	± 112 arcsec 3σ

Notes:
[1] The instrument provider should combine all static errors and dynamic errors separately then add the static and dynamic errors in reporting pointing accuracy and knowledge budgets for each axis, using the method identified in Figure 5.4.2-2. The static term for ground-to-on-orbit thermal shift should be broken into 2 pieces. The model predict, without a Model Uncertainty Factor (MUF), should be accommodated in the Pointing Accuracy budget and the uncertainty, with a MUF = 1.6, should remain in the Pointing Knowledge budget. Thermal distortion dynamic error terms should be treated together as a worst-case summation. A MUF = 1.6 should be used for all thermal distortion errors. The instrument provider is free to use a more conservative error combining method.

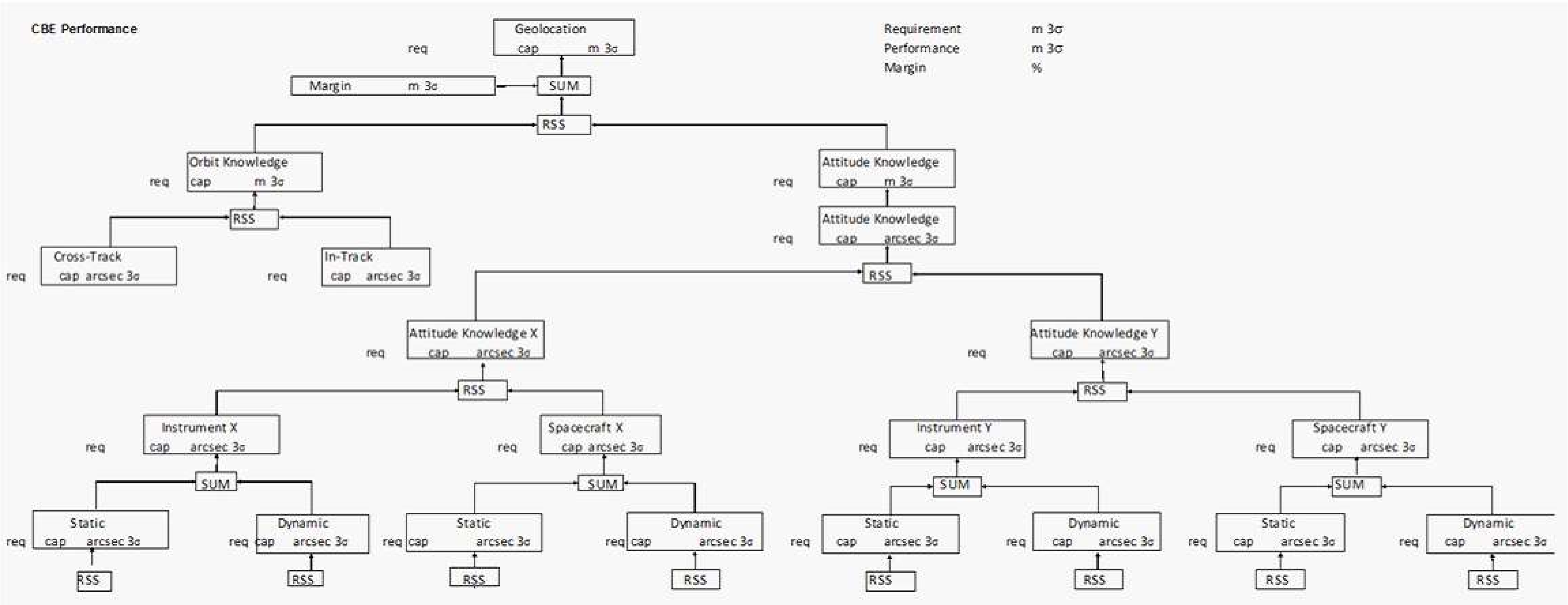

Figure 5.4.2-2. Analysis of Geolocation Knowledge Uncertainty at Nadir

5.4.3 Instrument Timing Requirements

The spacecraft will provide the instrument with a time-of-day (TOD) pulse and TOD packet that is accurate to +/- 1 millisecond of UTC to enable the instrument to synchronize a time reference for instrument data.

SMBA-116	The instrument shall use the TOD pulse and TOD packet to synchronize a time reference for instrument data with an accuracy of +/- 500 microseconds of the spacecraft time. Using this synchronization, any instrument packet time tag will be within +/- 1.5 milliseconds of UTC.

In event of brief loss of TOD pulses, both the instrument and spacecraft should continue in the current mode, including instrument processing of secondary header time stamp, until return of TOD pulses.

5.5 NEON Series-1 Project Mission Phases

The following section delineates requirements applied to the mission phases defined in NEON Series-1 Project Concept of Operations (ConOps) (NS1-xxxx-xxx-xxxx). The mission phases of the NEON Series-1 Project are as follows:

- Satellite Assembly, Integration and Test, and Ground Storage
- Pre-launch/Launch Site Processing
- Launch and Orbit Insertion Phase
- Satellite Activation, Checkout, and Commissioning Phase
- Mission Operations
- End-of-Life Decommissioning

5.5.1 Satellite Assembly, Integration and Test, and Ground Storage Phase

SMBA-117	The instrument shall be capable of remaining integrated to the spacecraft while undergoing testing at periodic intervals (e.g., 3-month periods) during assembly, integration and test (I&T), and ground storage.

SMBA-118	During assembly, cleaning, system I&T, and launch site processing, the instrument shall be capable of being processed in an ISO 14644-1 Class 8 cleanroom environment (or better).

NOTE: All instruments are expected to be covered and purged in a gaseous nitrogen whenever exposed to an environment worse than Class 8.

SMBA-119 Instrument Command and Telemetry Electrical Ground Service Equipment (EGSE) shall provide the following instrument Ground Service Equipment (GSE) interface functionality:

a. Transmit commands to the instrument.
b. Report the receipt, transmission, and execution of instrument commands by the instrument.
c. Receive instrument telemetry data in real-time.
d. Receive instrument science data in real-time.
e. Provide a simulated TOD pulse/message to the instrument.

5.5.2 Pre-Launch/Launch Site Processing Phase

During Pre-Launch/Launch Site Processing phase, the satellite is performing functional checkout of satellite (spacecraft and instrument) subsystems, integrating to the launch vehicle, and performing final mission readiness activities to support launch.

SMBA-120 The instrument shall be designed to perform satellite-level aliveness tests while mated to the launch vehicle. The aliveness test powers up the satellite and performs elementary power-on self-tests that gives assurance that the spacecraft and instruments will work as expected.

5.5.3 Launch and Orbit Insertion Phase

During Launch/Orbit Insertion phase, the satellite is performing functional checkout of spacecraft subsystems and maneuvering the satellite into the final mission orbit. During this phase, it is expected that the instrument is OFF, and its environment is being monitored/maintained by spacecraft-provided survival services.

5.5.4 Satellite Activation, Checkout, and Commissioning Phase

During the first 45 days after launch, the satellite will go through activation and checkout of all spacecraft subsystems and the instrument. This will include turn-ons, calibrations, outgassing, systems operations verifications, etc. The commissioning will culminate with a successful Post-Launch Assessment Review (PLAR) within 60 days following launch.

Note: While full commissioning is dependent on instrument contributions, the spacecraft is expected not to hinder these instrument contributions.

5.5.5 Mission Operations Phase

SMBA-121 The instrument shall maintain an operational availability of greater than 99.5% over any 30-day period for the mission lifetime. This includes the total time spent in planned instrument outages that prevent acquisition/transmission of operational data.

5.5.6 End-of-Life Decommissioning Phase

SMBA-122	The instrument shall limit orbital debris generation per NPR 8715.6, NASA Procedural Requirements for Limiting Orbital Debris and the companion document NASA-STD-8719.14, version C, Process for Limiting Orbital Debris.

5.6 Instrument Functional Requirements

5.6.1 Typical Satellite Modes

Figure 5.6.1-1 depicts the satellite mode and sub-mode configurations and mode transitions from pre-launch through on-orbit operations and decommissioning.

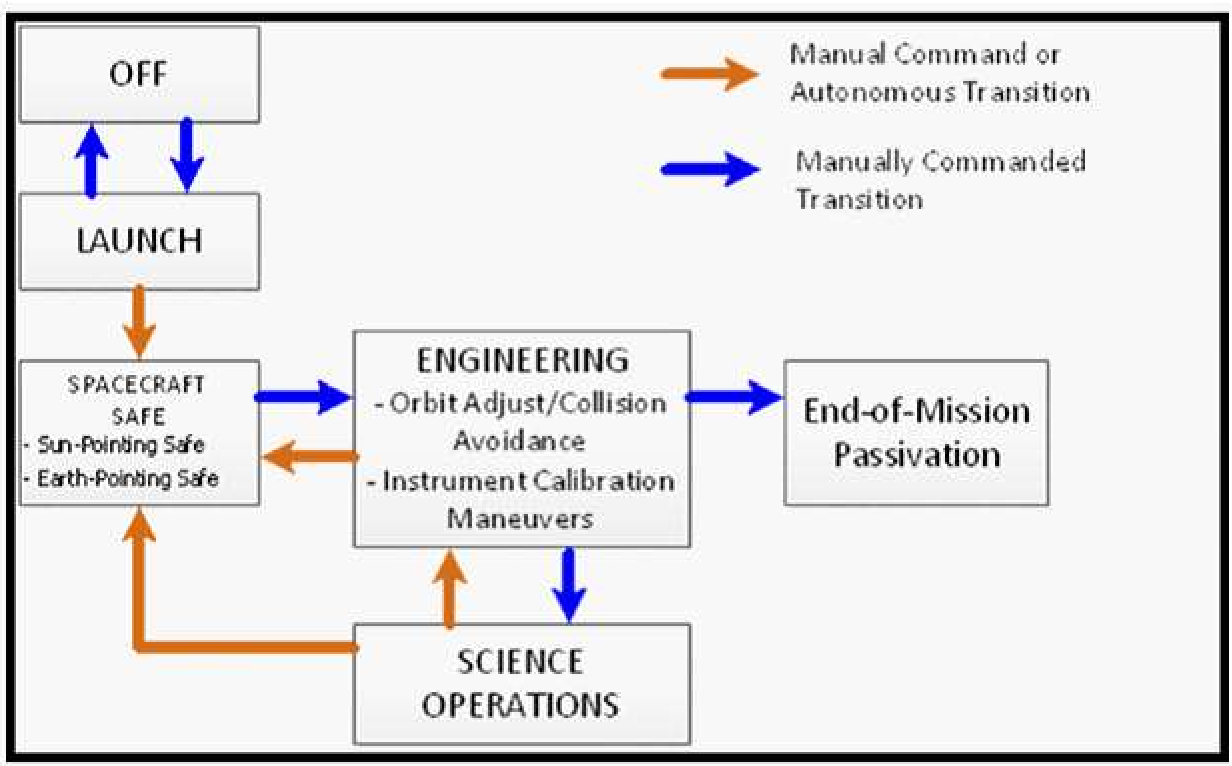

Figure 5.6.1-1. Typical Satellite Mode Transitions

The spacecraft should transition between modes by manual command and/or autonomously (as notionally depicted in Figure 5.6.1-1).

5.6.1.1 Spacecraft OFF Mode

The spacecraft should be able to be remotely configured into OFF Mode during ground operations. OFF Mode is defined as having all power removed including internal battery power.

5.6.1.2 Launch Mode Functional Requirements

The spacecraft should be capable of launching with the instrument in either Instrument OFF Mode or Instrument Survival Mode. Following orbit insertion and initial spacecraft deployments, the spacecraft will transition the instrument into Instrument Survival Mode.

5.6.1.3 Spacecraft Safe Modes

In the event of certain anomalous conditions, as determined by the spacecraft, the spacecraft will place itself and the instruments into a Safe Mode. Power system anomalies and attitude system anomalies are examples of conditions which could result in the entry into the Spacecraft Safe Mode. This mode could involve changing the nominal mission attitude of the spacecraft. This is an emergency mode and, as such, is a power conservation mode. As part of the process of entering Spacecraft Safe Mode, the spacecraft shall initiate the corresponding Instrument Safe Mode. For severe emergencies, the spacecraft may remove operational power to the instruments after ensuring the instrument survival heater services are enabled.

The spacecraft will provide a tiered approach to Spacecraft Safe Mode commensurate with the anomaly:

a. Earth-Pointing Safe Mode (as described in Section 5.6.1.3.1)
b. Sun-Pointing Safe Mode (as described in Section 5.6.1.3.2).

The spacecraft should be capable of acquiring the appropriate safe hold attitude from any arbitrary attitude while maintaining a safe environment for the spacecraft and instruments and without violation of instrument solar constraints. The spacecraft should limit exposure of the instruments to solar input onto sensitive surfaces, to the extent that sensitive surfaces and exposure limits are defined in the individual Spacecraft-to-instrument ICDs.

SMBA-123	The instrument shall be designed to meet all performance requirements in operational mode following the execution of either safing event described in sections 5.6.1.3.1 and 5.6.1.3.2.

5.6.1.3.1 Earth-Pointing Safe Mode

The spacecraft will provide a coarsely controlled Earth-Pointing Safe Mode capability that maintains a safe environment for the spacecraft and instruments for an indefinite period of time as the preferred mode for selected anomalous conditions. The Earth-Pointing Safe Mode attitude reference is defined as orienting the spacecraft body frame such that the spacecraft +Z-axis is aligned with geodetic or geocentric nadir and the +X axis is aligned with the projection of the instantaneous orbital velocity vector onto the plane perpendicular to geodetic or geocentric nadir. See illustration in Figure 5.6.1-2. (TBR)

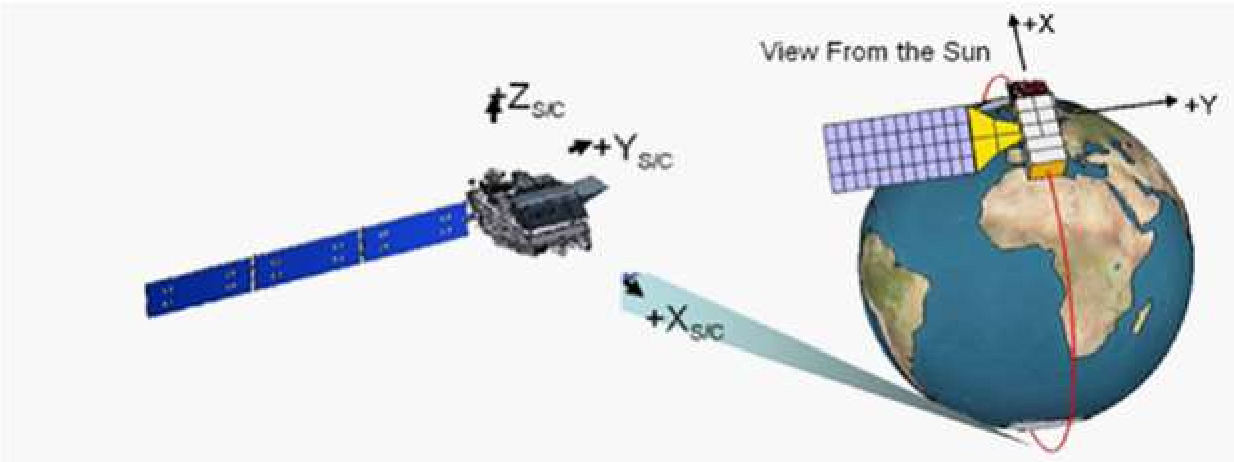

Figure 5.6.1-2. Typical Spacecraft Configuration for Earth-Pointing Safe Mode

5.6.1.3.2 Sun-Pointing Safe Mode

The spacecraft will provide a solar-inertial Sun-Pointing Safe Mode capability (see illustration in Figure 5.6.1-3) to point the solar array to the Sun while maintaining a safe environment for the spacecraft and instrument for an indefinite period of time.

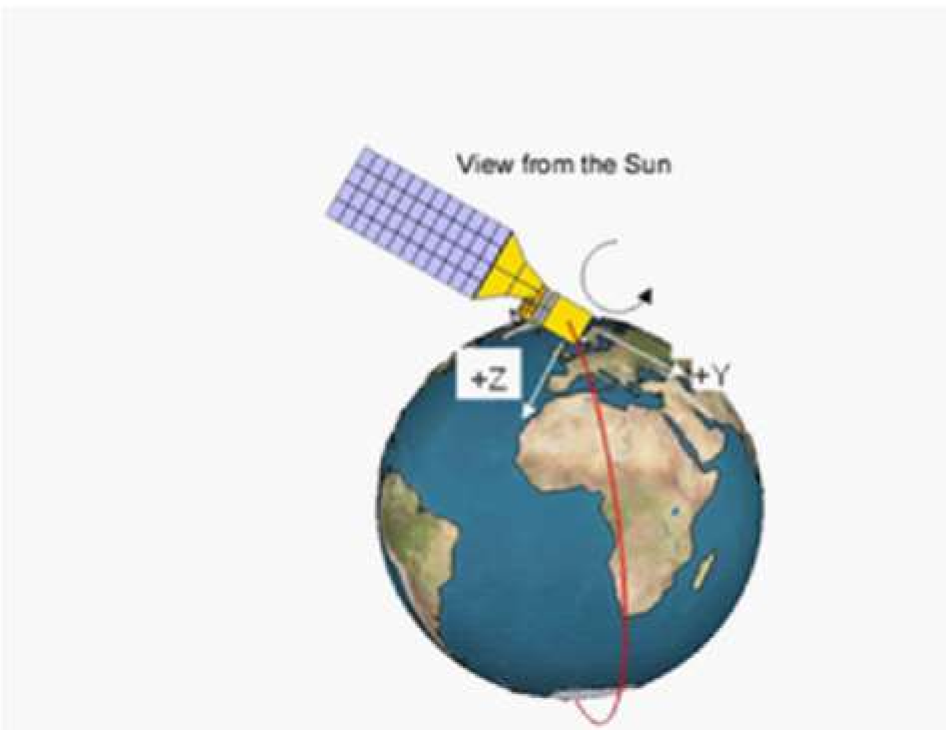

Figure 5.6.1-3. Typical Spacecraft Configuration for Sun-Pointing Safe Mode

5.6.1.4 Spacecraft Engineering Mode

Engineering Mode is the collection of modes where the spacecraft initially performs check-out, instrument outgassing, and other commissioning activities, as well as performs routine maintenance activities such as orbit adjust maneuvers and instrument sensor calibration maneuvers. While the attitude control requirements may be the same as in Science Operations Mode, time spent in Engineering Mode does not get counted towards satisfaction of the mission availability requirements with the exception of instrument calibration maneuvers.

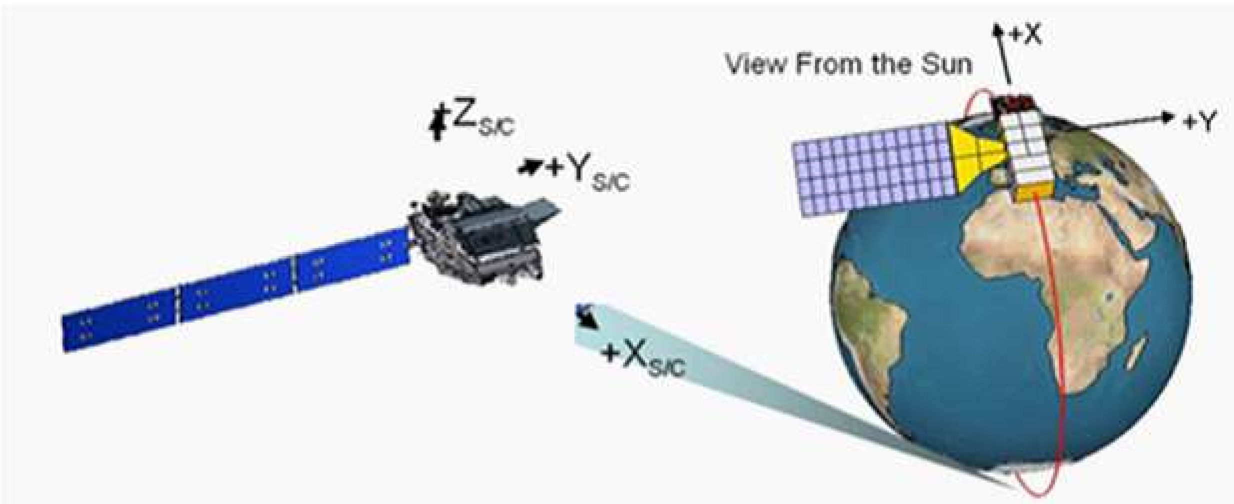

Figure 5.6.1-4. Typical Spacecraft Configuration for Engineering Mode and Science Operations Mode

During Engineering Mode and Science Operations Mode, the spacecraft will orient the spacecraft body frame such that the spacecraft +Z-axis is aligned with geodetic nadir and the +X axis is aligned with the projection of the instantaneous orbital velocity vector onto the plane perpendicular to geodetic nadir. See illustration in Figure 5.6.1-4.

During Engineering Mode and Science Operations Mode, the spacecraft will be capable of providing time-tagged spacecraft attitude estimates (Earth Centered Inertial (ECI)) to spacecraft Control Frame) and/ or a time-tagged spacecraft ephemeris (Earth Centered Earth Fixed (ECEF)) with a minimum frequency of 8 Hz (TBR) if required by the instrument to meet performance requirements.

SMBA-124	The instrument shall provide all data necessary to produce the raw data records used to create the science mission data while in either Engineering mode or Science Operational mode.

5.6.1.4.1 Orbit Adjust/Collision Avoidance

SMBA-125	The instrument shall be capable of remaining in Operational mode without damage while the spacecraft performs orbit correction.

5.6.1.4.2 Instrument Calibration Maneuvers

While in Spacecraft Engineering Mode, the spacecraft performs the calibration maneuvers upon ground command and returns to nominal satellite operations (including earth-pointing orientation) upon completion.

SMBA-126 The instrument shall have the capability of performing the calibration maneuvers listed in Table 5.6.1-5 (TBS) upon ground command, non-simultaneously, and returning to nominal satellite operations (including earth-pointing orientation) upon completion.

Table 5.6.1-5. Suggested Instrument Calibration Maneuver Requirements (TBS)

Maneuver	Description	Duration (time in final attitude)	Constraints	Rationale
SMBA Earth Limb Image	Once (during Commissioning) Roll (up to +38°)	Dwell for up to 5 minutes (4 minute minimum)	Eclipse, clear sky, over ocean, to allow main lobes of outer most beams to be scanned off the Earth limb	Measure: - Magnitude of cross-track scan bias - Magnitude of sidelobe contamination - Radiometric equivalency of Earth/Space-view
SMBA Deep Space (Back-flip)	Once (during Commissioning) Pitch (+360°) over ~⅓ orbit (Eclipse duration) @ slowest constant rate to return to Earth-pointing by Eclipse exit	~⅓ orbit (Eclipse duration) No Dwell at target Desired slew rate (~0.15 deg/sec)	Eclipse, starting at terminator crossing in the Science Mode attitude; return to Earth-pointing by Eclipse exit Perform 1-3 days after full moon (SMBA to view moon during backflip maneuver)	Primary: Perform on-orbit evaluation of beam-pointing and check for RFI interference from on-board transmitters. Secondary: Amplitude calibration, sidelobe characterization. Obtain knowledge of non-uniformities/biases across Earth view sector and estimate emissivity of optical chain elements.

Note: Calibration maneuvers listed are typical for on-orbit commissioning. Instrument provider should propose maneuvers necessary for successful calibration and science operations.

5.6.1.5 Spacecraft Science Operations Mode

Science Operations Mode is the predominant operating mode during the mission. All time spent in the Science Operations Mode counts towards the satisfaction of the mission availability requirements defined in Section 5.5.5. The spacecraft will provide a three-axis stabilized nadir pointing attitude control system capable of meeting the geolocation requirements of the science instruments defined in Section 5.4.1.

The spacecraft should take no more than 10 minutes to return from Engineering Mode or Earth-Pointing Safe Mode to Science Operations Mode.

The spacecraft should take no more than 30 minutes to return from Sun-Pointing Safe Mode to attitude control with the assumption that the instrument interface temperature conditions are capable of Science Operations Mode.

5.6.1.6 End-of-Mission Passivation Mode

At the end of the mission, the instrument will be powered off prior to the start of the disposal maneuvers.

5.6.2 Typical Instrument Modes

This section defines requirements associated with the five typical individual operating modes of the instrument. The typical five main instrument operating modes are:

- Instrument OFF Mode
- Instrument Survival Mode
- Instrument Safe Mode
- Instrument Engineering Mode
- Instrument Operational Mode

While the modes can be dependent on the spacecraft, when the spacecraft is in Operational (Science) Mode, the instrument can be in any of the above modes.

SMBA-127 The instrument shall implement the following modes and functionalities, as appropriate, with the mode characterizations shown in Table 5.6.2-1 and mode transitions shown in Figure 5.6.2-1.

Table 5.6.2-1. Instrument Mode Characterization

Instrument Mode	Survival Power	Operational Power	CMD & TLM	Meet Section 4 Science Performance
OFF	OFF	OFF	No	No
Survival	Enabled	OFF	No	No
Instrument Safe	Enabled[1]	ON	Yes	No
Engineering - Activation - Diagnostic	Enabled[1]	ON	Yes	No[2]
Operational	Enabled[1]	ON	Yes	Yes

Notes:
Yes = capability pertains to that mode/functionality
[1] Instrument Survival heater service may or may not be enabled, outside of Survival Mode.
[2] Science performance in Diagnostic mode is not precluded, but is not required

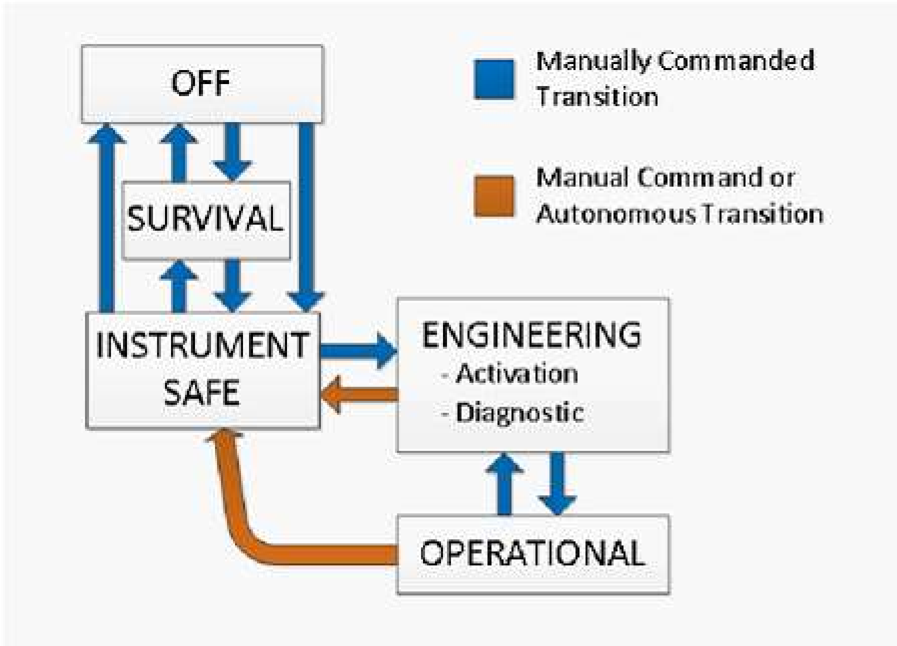

Figure 5.6.2-1. Typical Instrument Mode Transition Diagram

SMBA-128	The instrument shall be command-able into any of the instrument modes, with transitions as illustrated in Figure 5.6.2-1.

SMBA-129	Although preferred instrument-provided sequences may be defined, the instrument shall cause no damage to any instrument or the spacecraft when transitioning from any mode to any other mode.

5.6.2.1 Instrument OFF Mode

In the OFF Mode, no operational power is supplied to the instrument. In this mode, the spacecraft controls the instrument via the pulse command lines, rather than by the C&T bus. The spacecraft also monitors relay status indicators via the discrete bi-level telemetry interfaces.

Instrument OFF Mode may be used for ground storage & transportation, launch, and spacecraft power crisis situations.

Note: In the OFF Mode, the instrument will receive no power from the spacecraft via the operational or the survival power service.

5.6.2.2 Instrument Survival Mode

Survival Mode is entered when a critical power shortage has been identified by the spacecraft. Survival Mode is a low power mode from which the Observatory can eventually recover to full operational status. Only those functions required for Observatory safety, diagnostics, and recovery will be powered. Instrument operational power will be OFF and instrument survival power will be enabled. In Survival Mode, the spacecraft will be responsible for sampling critical instrument temperatures via the instrument passive analog temperature sensors (normal instrument telemetry is not available with operational power OFF). Initiation of Survival Mode by the spacecraft should not require commands to turn on the instrument survival heaters.

SMBA-130 In Survival Mode, the instrument shall be able to survive indefinitely while the spacecraft is in either safe mode attitude defined in section 5.6.1.3.

5.6.2.3 Instrument Safe Mode

In the event of certain on-orbit spacecraft bus subsystem anomalies, the instrument must be properly configured to protect against prolonged exposure to abnormal environments such as those caused by abnormal pointing of the spacecraft. When possible, the spacecraft will send pre-defined reconfiguration commands to the instrument specified in the Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx), including moving the scanning reflector to a stowed position. In addition, the instrument must put itself into a Safe Mode upon detection of an internal anomaly which could be detrimental to the instrument or any other part of the Observatory.

SMBA-131 For instances of an instrument-detected on-orbit failure, where failure to take prompt corrective action could result in damage to the instrument, the instrument shall autonomously place itself into a Safe Mode. Autonomous transition to Safe Mode may be inhibited by the on-orbit upload of safety limits for mission-critical parameters.

SMBA-132 The instrument shall enter Safe Mode upon receipt of a spacecraft command to enter Safe Mode.

SMBA-133 The instrument shall autonomously configure to its Safe Mode within 45 seconds after entry into Safe Mode is initiated.

SMBA-134 The instrument shall provide a mode status flag in housekeeping telemetry, to indicate when it is in Safe Mode.

SMBA-135 Upon initiation of Safe Mode, the instrument shall configure itself such that no damage will occur if the next action from the spacecraft is to turn OFF the instrument, even though that may not be the normal flow.

5.6.2.4 Instrument Engineering Modes

Instrument Engineering Modes are the collection of modes where the instrument initially performs the post-launch check-out and commissioning activities such as activation, outgassing, and calibrations, as well as performs routine maintenance activities such as software updates, outgassing, and sensor calibrations. While the science performance may be the same as in Operational Mode, time spent in Engineering Modes does not get counted towards satisfaction of the mission availability requirements defined in Section 5.5.5 with exception of science calibration maneuvers.

5.6.2.4.1 Instrument Activation Mode

Activation Mode refers to instrument turn-on, and subsequent instrument component warm up, or cool down, to the operating temperatures. Activation Mode terminates when all instrument temperatures, biases, and currents have stabilized within specified operational limits.

SMBA-136	During power up, the instrument shall enable the command and telemetry functions and provide for orderly turn-on (channel enable, scan motor turn-on, etc.) based on commands to the instrument.

SMBA-137	Following power-up, the instrument shall be functional without requiring a software or data upload.

5.6.2.4.2 Instrument Diagnostic Mode

Diagnostic Mode refers to instrument engineering activities (diagnostic procedures) to evaluate science performance, update software tables/configurations, and/or collect status/health telemetry.

SMBA-138	To support early orbit checkout, and for anomaly resolution, the instrument contractor shall provide the capability to selectively disable any on-orbit processing operation that combines or compresses raw data in any manner.

SMBA-139	Instrument Diagnostic mode shall support housekeeping, telemetry and software updates, as well as troubleshooting activities including capability of different sampling of telemetry (rates and memory locations).

SMBA-140	In Diagnostic Mode, the instrument shall be capable of collecting, storing, and relaying to ground raw data used for RFI detection. Note: Raw data refers to internal instrument data along the processing chain prior to the formation of channels. Raw data may include data in time domain and/or frequency domain.

In Diagnostic Mode, the instrument can point the scan reflector to any commanded scan reflector position. The commanded position is maintained until a new pointing or scanning command is received.

SMBA-141	In Diagnostic Mode when commanded, the instrument shall scan and position the antenna beam electrical boresight to enable measurement of scene radiance at specified locations.

SMBA-142	In Diagnostic Mode, the instrument shall have the capability to provide continuous sampling of science data.

SMBA-143	In Diagnostic Mode, the instrument shall have the capability to scan at a constant velocity.

During ground testing, the instrument should have the capability to use RFI test cases developed collaboratively between NASA and the chosen instrument provider. This capability allows for RFI detection algorithm performance to be assessed. RFI test case data can be imported via test port, simulated in the lab environment, and/or commanded via built-in test. The instrument should provide insight along the processing stream from end-to-end for validating RFI detection algorithm performance and radiometric performance.

The instrument should have the capability to perform autonomous and commanded built-in tests. Autonomous built-in tests are performed in the background to monitor instrument health and safety in order to support fault identification, in both operational and diagnostic modes. Commanded built-in tests are performed only when commanded in the diagnostic mode. Table 5.6.2-2 provides suggestions for built-in tests.

Table 5.6.2-2. Built-in Tests

(This table can be revised to reflect the specifics of the instrument chosen)

Test	Description	Built-in Test type
Safety	Monitor key voltages and temperatures against safety limits	Autonomous
ADC	Monitor reference voltage injected into multiplexer/ADC	Autonomous
Memory	Memory test performed as part of boot-up	Commanded
Dwell test	Sample a specific housekeeping channel at maximum acquisition rate and provide data back to the ground	Commanded
Point & stare	Continuous sampling while pointing scanner to a selected beam position	Commanded
Constant Velocity Scan	Continuous sampling while scanning at constant velocity	Commanded

5.6.2.5 Instrument Operational Mode

Instrument Operational Mode is the predominant operating mode during the mission. All time spent in this mode counts towards the satisfaction of the mission availability requirements defined in section 5.5.5.

SMBA-144	In the Instrument Operational Mode, the instrument shall be fully functional, providing all data originating within the instrument to produce the raw data records (RDRs) as well as health and status telemetry, RFI detection, and housekeeping data.

SMBA-145	The instrument shall be capable of remaining in Operational Mode without damage while the spacecraft performs orbit maintenance maneuvers.

5.6.3 Contingency Management

The instrument should be capable of indefinitely maintaining its health and safety without ground support.

SMBA-146	The instrument shall perform autonomous fault detection, isolation, and correction for mission-critical functions where faults could cause damage, loss of control, or loss of science.

SMBA-147	The instrument shall employ telemetry monitoring (TMONS) or similar capability to aid in the detection and reporting of out-of-limit conditions or other abnormal affects that could be caused by external sources of Radio-Frequency Interference (RFI) within 1 orbit of the occurrence.

SMBA-148	An unresolvable instrument subsystem fault shall result in autonomous transition to Instrument Safe Mode.

5.7 Instrument Design Requirements

The following sections delineate the design requirements and guidelines of the various instrument technical disciplines taken from applicable NASA and industry design standards.

5.7.1 Mechanical Alignment Requirements

A permanently affixed optical alignment target should be used to reference antenna pattern measurements and provide the spacecraft integrator a known relative indicator of boresight alignment.

SMBA-149	The alignment of the instrument beam electrical boresight axes shall be measured with respect to the primary optical alignment target.

SMBA-150	The alignment of the optical alignment targets with respect to the antenna subsystem mounting surfaces shall change by less than or equal to +/-0.025 degrees as a result of any testing.

SMBA-151	The instrument alignment reference shall be viewable from two orthogonal directions when integrated to the satellite.

5.7.2 Structural Requirements

The instrument will survive the critical loading conditions that exist within the envelope of handling, launch environment, and mission requirements. The detailed requirements are provided in the following section.

SMBA-152	Instrument structure(s) shall be designed to have positive margin of safety for all failure modes to the worst simultaneous combination of design limit loads, applied temperature, and other accompanying environmental phenomena for each design condition without experiencing yielding or detrimental deformation.

Limit load is defined as the highest expected load, including environmental effects. Limit loads due to the launch environment or other acceleration environment will be applied through the center-of-mass of the components, as configured for the appropriate instrument mode, to design the component mounting interfaces.

SMBA-153 Instrument structure(s) shall be designed to have positive margin of safety for all failure modes to the worst case expected launch load of 15 g independently in each axis (TBR).

The factors of safety shown in Table 5.7.2-1, or those defined in the notes to the table, as applicable to the type of calculation listed in the table, will be applied to limit loads to calculate structural margins.

Table 5.7.2-1. Flight Hardware Design/Analysis Factors of Safety Applied to Limit Loads [1] [2]

Type	Static	Sine	Random/Acoustic[4]
Metallic Yield	1.25[3]	1.25	1.6
Metallic Ultimate	1.4[3]	1.4	1.8
Stability Ultimate	1.4	1.4	1.8
Beryllium Yield	1.4	1.4	1.8
Beryllium Ultimate	1.6	1.6	2.0
Composite Ultimate	1.5	1.5	1.9
Bonded Inserts/Joints Ultimate	1.5	1.5	1.9

Notes:
[1] Factors of safety for pressurized systems to be compliant with NASA-STD-8719.24.
[2] Factors of safety for glass and structural glass bonds specified in NASA-STD-5001, Tables 4 and 5.
[3] If qualified by analysis only, positive margin will be shown for factors of safety of 2.0 on yield and 2.6 on ultimate.
[4] Factors shown should be applied to statistically derived peak response based on root mean square level. As a minimum, the peak response will be calculated as a 3-sigma value.

5.7.3 Mechanism Design Requirements

The following design requirements apply to the mechanisms used in the design of the instrument. NASA-STD-5017 Mechanism Design Requirements provides suggested best practices for mechanism designs.

5.7.3.1 Torque/Force Margins

SMBA-154 The minimum torque at the motor shaft shall never be less than 7.06E-3 N-m (1 oz-in).

Torque margin is a measure of the excess capability of a mechanism to impart, maintain, or prevent motion, or to provide required acceleration. Torque margin is intended to ensure that a mechanism retains reserve torque that can be applied in the event of an unforeseen effect that increases resistive torque or reduces motive torque within the mechanism, similar to the factor of safety used in a structural margin calculation.

Torque margin is not applicable to mechanisms that must provide a specific torque value within a narrow tolerance range to perform their basic function, such as a spring holding a relief valve closed or an ejection mechanism requiring a specific (not minimum) separation velocity.

Driving Torques: T_{avail} is the minimum available torque generated by the driving or holding component (e.g., spring, motor, pyrotechnics, solenoids, heat-actuated devices, pneumatic or hydraulic systems, brake). The minimum available torque from an energized motor is typically understood to include "magnetic losses" such as hysteresis and eddy current drag torques, "mechanical losses" such as bearing, brush, and windage drag torques, and "torque ripple" caused by detent (cogging) torque, asymmetry in motor design and construction, commutation strategy, etc.

T_k are the individual maximum known resisting torques that are not strongly influenced by environmental conditions and cycles.

For static and dynamic torque margins, T_v terms are the individual maximum "variable" resisting torques whose values may change with environmental conditions and cycles (e.g., friction torque, viscous drag torques, wire harness torque due to flexing or set). For holding torque margins, T_v is used to denote the individual maximum disturbing torques regardless of their degree of variability.

T_f are the individual maximum "fixed" resisting torques that are not strongly influenced by environmental conditions and cycles (e.g., vehicle maneuver-induced torques, return spring torques, unbalanced pressure loads limited by relief mechanisms).

T_a is the torque required to achieve a specified acceleration of the driven component.

Resistive Torques: Sum of Torque known = sum of the fixed torques or forces that are known and quantifiable such as accelerated inertias ($T=I_a$) and not influenced by friction, temperature, life, etc. A constant FS is applied to the calculated torque required. Sum of Torque variable = sum of the torques or forces that may vary over environmental conditions and life such as static or dynamic friction, alignment effects, latching forces, wire harness loads, damper drag, variations in lubricant viscosity, including degradation or depletion of lubricant over life.

There are many forms of torque margin equations in use in various standards. Each can be reformulated to appear like the others; the only difference among them is in the magnitude and nature of the different safety factors applied to the terms. This form was chosen due to its relative simplicity, its ability to handle several torque margin calculations with a single equation, and its suitability for application of maturity-based safety factors.

SMBA-155	Sufficient torque or force margins shall be provided to assure reliable mechanism operation over the life of the mission. a. Margins to be determined using worst-case conditions and include all flight drive electronics effects and limitations. b. Torque margin to be verified using measured data acquired from unit qualification for a new design. Subsequent units to be verified using acceptance level operational test data. c. The Torque Margin (TM) to be greater than zero and be calculated using the following formula: $TM = \{T_{avail} / (FS_k\ T_k + FS_v\ T_v + FS_a\ T_a)\} - 1$ d. The minimum available driving torque for the mechanism to be determined based on the factor of safety (FS) defined in SMBA-156.

SMBA-156	The Mechanism Design Factor of Safety shall be 1.15 for Acceleration Torques (FS_a), 1.35 for Known Torques (FS_k), and 2.5 (FS_v) for Variable Torques. Note: Reference NASA-STD-5017B for further clarification.

5.7.3.2 Binding/Jamming/Seizing

SMBA-157	Mechanisms shall either be designed to preclude installation in an incorrect orientation or else be clearly labeled in a manner that indicates proper installation orientation and prevents improper installation.

Designs should include provisions to prevent binding, jamming, or seizing.

Examples of possible provisions include dual rotating surfaces or other mechanical redundancies; robust strength margins such that self-generated internal particles are precluded; shrouding and debris shielding; proper dimensioning and tolerancing; thermal analysis; proper selection of materials and lubrication design to prevent friction welding, galling, etc.; and testing to determine detrimental contamination levels.

5.7.3.2.1 Clearances

SMBA-158	Static and dynamic clearances between the mechanism and any other structure, component, thermal covering, and field of view shall be established and maintained, during all phases of the mission at the highest level of assembly. Note: NASA-STD-5017B, Section 4.2 provides guidance on best practices to maintain static and dynamics clearances.

Maintaining clearances within and around mechanisms is necessary both to maintain proper mechanism function and to prevent the mechanism from causing problems with other systems. Establishing clearances allows one to both design the mechanism to maintain the clearances and then verify the existence of those clearances in the design and on the flight hardware.

Established clearances should account for the following:

a. Manufacturing, assembly, and alignment tolerances.
b. Temperature- and temperature-gradient-induced deformations or instabilities.

c. Deflections due to vibration.
d. Deflections due to external loads, including gravity effects.
e. Deflections due to operational loads.
f. Deflections due to changes in pressurization, including thermal blanket billowing.
g. Motion of cable harnesses, tubing, and sensor wiring.
h. Environments arising from transportation.
i. The full range of adjustability of the mechanism parts.

5.7.3.2.2 Tolerancing

SMBA-159 Dimensional tolerances on all moving parts and intentional interference-fit parts shall be established and documented via a tolerance stack-up/clearance analysis to ensure that proper functional performance is maintained under all natural and induced environmental conditions and configurations.

Natural and induced environmental conditions and configurations include but are not limited to: (1) thermally induced in-plane distortions to include, differential thermal growth and shrinkage, (2) deflections due to internal and externally applied loads, and (3) mechanical adjustment (rigging).

5.7.3.2.3 Lubrication

5.7.3.2.3.1 Lubricant Compatibility

SMBA-160 Lubricants used in the mechanism shall be compatible with the following: (1) interfacing materials (including components and fluids), (2) other lubricants used in the mechanism, (3) all natural and induced environments encountered by the mechanism, (4) outgassing/creep requirements (e.g., for nearby optical surfaces), if applicable, and (5) hygroscopic requirements, if applicable.

5.7.3.2.3.2 Lubricant Life

SMBA-161 The selection of lubricant for use in the mechanism shall be based upon development tests of the lubricant that demonstrate its ability to provide adequate lubrication under all specified operating conditions over the design lifetime.

5.7.3.2.3.3 Bearing Lubrication

SMBA-162 The selection of lubricant for use in critical moving mechanical assemblies shall be based upon development tests or flight heritage of the lubricant that demonstrate its ability to provide adequate lubrication under all specified operating conditions over the design lifetime.

SMBA-163	An analysis shall be performed for any liquid lubricant application, subject to depletion, to show that there is an adequate amount of lubricant in the system (not including degradation) for the duration of the mechanism's operational life with a factor greater than 10.

If solid lubrications are used, specific written procedures should control the method of application, subsequent handling, and ground testing of the components and assemblies to avoid exposure to moisture or humidity.

5.7.3.3 Scan Motor

SMBA-164	The torque capability of the scan drive motor shall be sufficient to perform its proper functions when the frictional torque is three times the predicted worst case.

SMBA-165	The predicted worst-case frictional torque shall include worst-case combinations of environment (particularly temperature extremes), tolerances, lubricant contamination and/or evaporation, and other possible degradation at the end of the specified operational period.

5.7.3.3.1 Scan Control Capability

SMBA-166	The scan position encoder shall provide 360-degree position readout.

SMBA-167	The beam scan angle of every beam position shall be digitized to an accuracy of no fewer than 16 bits.

SMBA-168	If the digitizer selected is an optical encoder, filament type shall not be utilized.

SMBA-169	The scan drive control loop shall have stiffness adequate to achieve specified pointing performance in the presence of at least three times the predicted worst-case frictional torque.

SMBA-170	The scan drive control loop shall have an open loop phase margin no less than 25 degrees.

SMBA-171	The scan drive control loop shall have an open loop gain margin no less than 12 dB within ±10% of structural mode frequencies, and no less than 6 dB at other frequencies. (TBR)

SMBA-172	The scan drive subsystem shall be analyzed and tested to ensure that scan control characteristics meet the overall system pointing error budget.

5.7.3.3.2 Contingency Capability

SMBA-173	The scan drive motor shall meet all performance requirements in operational mode after being powered and commanded to start up under locked rotor conditions, in an air or vacuum environment for 30 minutes.

SMBA-174	The scan drive mechanism shall meet all performance requirements in operational mode following any available shutdown mode or sequence.

5.7.3.4 Mechanism Performance and Strength Analysis

SMBA-175	Mechanism components and linkages shall have sufficient strength to tolerate an actuation force/torque stall condition at any point of travel and still maintain a positive margin of safety with the ultimate factor of safety applied.

SMBA-176	Bearings shall have analysis demonstrating acceptable material, mounting, preload, performance, and structural integrity, accounting for the following conditions: (1) maximum combined axial, radial, and moment loads sustained during ground handling, launch, on-orbit, or other operational modes, (2) system stiffness requirements, (3) effects of temperature, temperature gradients, fits, tolerances and initial preload on torque, stiffness, and life, (4) lubrication, (5) wear, (6) smoothness of operation, (7) friction torque, considering breakaway and running, in the installed state, (8) reliability and life, and (9) effects of alignments, fits, tolerances, thermal, and load-induced distortions on preload, stress, and bearing shoulder height requirements.

5.7.4 Electrical Systems Requirements

The spacecraft will provide electrical power generation, power management, power distribution, energy storage, load shedding control, battery state-of-charge management, and protection against over-voltage, under-voltage, and over-current during the mission lifetime in all mission modes.

SMBA-177	The instrument shall be designed to operate over a range of 26 volts direct current (VDC) to 34 VDC at the load interface.

SMBA-178	The instrument shall be designed to survive voltages from +0V up to +40 VDC at any instrument operational or survival power interface.

5.7.5 Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) Requirements

EMC is necessary in each of the following areas:

a. self-compatibility of an instrument to itself;
b. compatibility between an instrument and all internal sources and receivers on the satellite (other instruments and the spacecraft bus); and

c. compatibility between an instrument and all external sources and receivers "visible" on the ground, on the launch vehicle, and on-orbit.

While operating in the configurations necessary for transport, test, launch, separation, and on-orbit mission activities, the spacecraft bus, instruments, ground support equipment, and test equipment will operate together without performance degradation due to EMI from each other, or the external environment, and without interfering with equipment in the external environment.

SMBA-179	The instrument shall be designed to limit conducted and radiated emissions as specified by the EMI/EMC emission test limits defined in Section 7.3.

SMBA-180	The instrument shall be designed to meet performance specifications when subjected to the conducted and radiated susceptibility tests defined in Section 7.3 as well as the On-Orbit RF environment in Section 6.2.

All connectors that are not used for flight should have flight-quality EMI-preventative covers that are captive when installed for launch.

5.7.6 Thermal Control Requirements

The instrument provider will work with the spacecraft vendor to define and document the thermal interface requirements in the ICD.

SMBA-181	The instrument shall be thermally safe for continuous operation in all spacecraft and instrument modes.

SMBA-182	The instrument shall be capable of maintaining instrument subsystems within their operational design temperature range when the spacecraft is in Science Operations Mode or Engineering Mode.

SMBA-183	The instrument shall be capable of maintaining instrument subsystems within their design survival temperature range when the instrument is in Safe or Survival Mode.

5.7.7 Contamination Control Requirements

Preferred contamination control processes are detailed in the JPSS Flight Project Contamination Control Plan (472-00228). Should there be any conflicts between the practices listed below and the contents of the Contamination Control Plan, this document has precedence.

Highlights include:

- The cleanliness levels specified are all updated per IEST-STD-CC1246E.
- The units of the cleanliness specification should be percent area coverage (PAC) for particulates and micrograms per square centimeter (µg/cm2) for molecular or non-volatile residue.
- Outgassing is specified as a total rate allocation with the units of grams per second (g/s).

- Internal surfaces should be cleaned and visibly inspected to VC-0.5-1000 per IEST-STD-CC1246E, Annex D prior to closeout or loss of access.
- External surfaces should be maintained to < 0.241PAC and < 1.00 µg/cm2, as defined in IEST-STD-CC1246E, prior to instrument integration.
- External surfaces should be maintained to < 0.241PAC and < 1.00 µg/cm2, as defined in IEST-STD-CC1246E, prior to satellite encapsulation in launch vehicle fairing.
- Non-metallic materials used in the construction of spacecraft components should not suffer a total mass loss of greater than 1.0 percent and a volatile condensable material mass gain of greater than 0.1 percent when tested in accordance with the Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment, ASTM E595. Higher values are permissible if subassemblies that contain these non-metallic materials meet their vacuum bakeout requirements.
- Materials for the spacecraft should be selected for low outgassing, using "Outgassing Data for Selecting Spacecraft Materials" [http://outgassing.nasa.gov] as a guide, and for resistance to the effects of incident radiation.
- Vents and outgassing paths should be controlled to minimize the extent to which molecular outgassing products have access to exterior surfaces and to vent outgassed species away from instrument critical surfaces.
- Instrument thermal control surfaces should be cleanable to visibly clean or better.
- Instrument thermal blankets should be cleaned and baked prior to installation.

5.7.8 Instrument Flight Software Requirements

SMBA-184 Flight processors providing computing resources shall be sized for worst-case utilization not to exceed the capacity shown in Table 5.7.8-1 (measured as a percentage of total available resource capacity):

Table 5.7.8-1. Flight Processor Resource Utilization Limits

Test Method	Measured
RAM	70%
PROM	100%
EEPROM	70%
CPU	70%

SMBA-185 On power-up or upon reset, the on-board computer (OBC) shall execute flight software after completion of the power-on-self-test and bootstrap process.

SMBA-186 The OBC shall be reset on execution of a discrete hardware command, or on execution of a flight software-initiated instruction or sequence of instructions.

SMBA-187	Execution of a soft reset shall not modify the contents of the data random access memory (RAM); except for pointers, indexes and flags intentionally reset by the soft reset, electrically erasable programmable read-only memory (EEPROM) and SDE RAM.

SMBA-188	The flight software shall be deterministic in terms of scheduling and prioritization of critical processing tasks to ensure their timely completion.

SMBA-189	The instrument shall have the capability to update flight software through ground commanding.

5.7.9 Instrument Simulator and Emulator Requirements

NASA, NOAA, and the NEON Series-1 spacecraft provider will need to simulate instrument functions for a variety of purposes. As directed in the NEON Series-1 SMBA Instrument SOW (NS1-SMBA-SOW-0002), the instrument provider will determine the proper combinations of software and hardware to meet the requirements of this section.

The spacecraft instrument interface simulator is a special category of EGSE. It will be used by the both the spacecraft and instrument developers to verify each electrical interface before instrument delivery to the spacecraft.

SMBA-190	The fidelity of the Spacecraft Instrument Interface Simulator (SIIS) shall be sufficient to validate all electrical interfaces between the spacecraft and the instrument: power bus; data bus for command, housekeeping telemetry, and high-rate science data; timing bus; and discrete analog and digital services.

6 SPACE ENVIRONMENT REQUIREMENTS

Specified below are natural environment characteristics in the presence of which the instrument will meet all other performance requirements.

6.1 General Radiation

The instrument will be compatible with the natural radiation environments for its operational orbit. To envelope the needs over the potential operational periods for the instrument, the document defines the radiation environment for the evaluation of degradation due to total ionizing and non-ionizing dose and of single event effects for the worst case sixty-two (62) month period for the NEON Program's missions spanning from January 2028 to December 2043.

SMBA-191 The instrument shall be capable of meeting all performance requirements at the total ionizing and non-ionizing dose levels and single event effect levels defined as specified in Radiation Environment for the NEON Program (470-LEO-ANYS-00018).

Total Ionizing Dose (TID):

- For nominal 2.5 mm (~100 mils) of equivalent aluminum shielding and the 62-month mission, the expected dose is 35.4 krad-Si (95% confidence).
- It is recommended that the 95% confidence level be used as a total ionizing dose requirement for EEE parts.

6.2 EMI/EMC - RF Environment

SMBA-192 Instrument components shall operate without performance degradation while exposed to the field while in the intended operational mode for the "On-Orbit External" and "On-Orbit Steady State" environments defined in the RS103 Table 6.2-1.

SMBA-193 Instrument components shall survive after being exposed to the field while in the intended mode for the "Launch Site/PPF" (Payload Processing Facility) environments defined in the RS103 Table 6.2-1, including the exceptions defined in the table notes.

SMBA-194 Instrument components powered at launch (not expected) shall operate without performance degradation while exposed to the field while in the intended operational mode for the "Launch Mode/Ascent" state environment defined in the RS103 Table 6.2-1.

Note: See Section 7.3 for RS103 Test Method details.

Table 6.2-1. Radiated Susceptibility Levels Due to Factory/Transport, Launch Site, Launch Vehicle, Ascent, and On-Orbit Phases (TBR)

Frequency	Launch Site/PPF (V/m) [2,3]	Launch Mode/Ascent (V/m) [2,3]	On-Orbit External (V/m) [5]	On-Orbit Steady State (V/m) [1]
10 kHz - 2 MHz	20	20	20	20
2 MHz - 30 MHz	20	20	20	20
30 MHz - 200 MHz	20	20	20	20
200 MHz - 420 MHz	20	20	20	20
420 MHz - 470 MHz	50	50	30	20
470 MHz - 1 GHz	30	30	20	20
1 GHz - 1.67 GHz	100	40	30	20
1.67 GHz - 1.71 GHz	100	40	30	20
1.71 GHz - 2.2 GHz	100	40	30	20
2.2 GHz - 2.3 GHz	200	72	30	36
2.3 GHz - 2.5 GHz	200	72	45	20
2.5 GHz - 5.25 GHz	200	72	40	20
5.25 GHz - 5.925 GHz	200	120	60	20
5.925 GHz - 7.25 GHz	50	40	40	20
7.25 GHz - 8.5 GHz	50	45	40	45
8.5 GHz – 10.6 GHz	200	60	60	20
10.6 GHz - 12.8 GHz	20	20	20	20
12.8 GHz - 14 GHz	100	20	20	20
14 GHz - 20 GHz	20	20	20	20
20 GHz - 24 GHz [4]	20	20	[7]	[7]
24 GHz - 25.5 GHz	20	20	[7] TBR	[7] TBR
25.5 GHz - 27.0 GHz	50	20	1.0	1.0, 50 [6]
27.0 GHz - 31 GHz	20	50	[7] TBR	[7] TBR
31 GHz - 33 GHz [4]	20	20	[7]	[7]
33 GHz - 40 GHz	20	20	TBS	TBS
40 GHz - 50 GHz	20	20	TBS	TBS
50 GHz - 60 GHz [4]	20	20	[7]	[7]

Notes:

[1] The On-Orbit Steady State (RS103) values are the maximum levels at any instrument and include ~6 dB EMI susceptibility margin. Values that exceed 20 V/m are due to the local transmitters. Individual instrument environments may be lower. See Spacecraft-to-SMBA Instrument ICD (NS1-SMBA-xxx-xxxx).

[2] The large values (100-200 V/m) do not apply across the entire frequency ranges shown, but at discrete frequencies within those bands. Consult NASA NEON for further information. The 'Launch Site/PPF' environments may be reduced by any combination of procedures, facility shielding, or shipping container shielding.

[3] The following survival test ranges are to be analyzed for compliance but not tested on the NEON Flight Units: "Launch Site/PPF" ≥ 100 V/m, "Ascent" ≥ 50 V/m.

[4] Frequencies 21.3 – 24 GHz, 31 – 33 GHz and 50 – 60 GHz may be analyzed for compliance in lieu of testing.

[5] On-Orbit External Modulation - 100µs pulses at 1 kHz for frequencies above 1 GHz.

[6] Applies at angles greater than 15 (TBR) degrees from the SMBA boresight.

[7] Levels are as specified by Table 6.2-2.

Table 6.2-2. Radiated Susceptibility Levels Due to In-Band Sources in On-Orbit Phases (TBR/TBS)

Frequency (GHz)		**Sensitivity (dBµV/m)**	
From	To	Boresight (On-Orbit External)	>15 degrees (On-Orbit Steady State)
20.00	21.20	TBS	TBS
21.20	21.40	TBS	TBS
21.40	22.21	TBS	TBS
22.21	22.50	TBS	TBS
22.50	23.57	TBS	TBS
23.57	54.03	2.5	45
24.03	24.80	42	85
24.80	25.50	64	106
27.04	28.90	TBS	TBS
28.90	30.40	82	132
30.40	31.24	43	92
31.24	31.56	3.0	52
31.56	32.40	43	92
32.40	33.00	82	132
50.00	50.20	TBS	TBS
50.20	50.40	TBS	TBS
50.40	52.60	TBS	TBS
52.60	59.30	TBS	TBS
59.30	60.00	TBS	TBS

7 VERIFICATION REQUIREMENTS

The following paragraphs delineate the functional performance tests and demonstrations to be conducted on the instrument and its subsystems.

7.1 Mission Requirements Verification

A comprehensive verification program as specified will be conducted to ensure that the instrument, instrument components, instrument subsystems, interface with the spacecraft, and ground support systems meet the specified requirements. Certain testing will be conducted at the instrument provider facilities, at the spacecraft provider facilities, during instrument level test, instrument integration onto the spacecraft, as well as at the integrated satellite level.

The baseline instrument verification test sequence, both at the instrument provider and spacecraft provider locations, is defined in the System Engineering section of the NEON Series-1 SMBA Instrument SOW (NS1-SMBA-SOW-0002).

All instrument qualification, protoflight, and acceptance testing should be done in accordance with GSFC-STD-7000B as specified and tailored in this document. In event of conflicts, the requirements of this specification will take precedence.

All instrument qualification and protoflight tests will be conducted with hardware of the final design that has passed the in-process production screens.

Test levels including any requests for tailoring or notching will be submitted to NASA for review prior to testing.

7.2 Electrical Functional Test Requirements

The following paragraphs delineate the functional performance tests and demonstrations to be conducted on the NEON Series-1 Satellite and its components, subsystems, and instruments.

7.2.1 Electrical Interface Testing

SMBA-195	As a part of the integration of a component or subsystem into the next higher level of assembly, electrical tests shall be performed to verify the interface configuration (power, grounds, commands, telemetry, signals, timing, etc.).

During the initial integration of the instrument on the spacecraft, turn-on and turn-off transients will be measured and characterized to ensure that equipment inrush current due to turn-on transients will neither adversely affect the power bus nor overstress its circuit protection device. Refer to the NEON Series-1 Spacecraft-to-SMBA instrument ICD (NS1-SMBA-xxx-xxxx) for the instrument inrush current limits.

7.2.2 Comprehensive Performance Tests (CPTs)

SMBA-196 Prior to mating with other hardware, electrical harnessing shall be tested to verify the wire routing, isolation, impedance, and overall workmanship. The following parameters shall be verified as minimum:

a. Accuracy (signals on correct pins and nowhere else).
b. Inputs and outputs (unloaded and loaded).
c. Specified range (high/low extremes as well as nominal).
d. Range impacts (how range extremes of one signal affect related signals).

SMBA-197 Comprehensive Performance Tests (CPTs) shall be conducted at the subsystem, instrument, and satellite levels of assembly to demonstrate that the hardware and software meet their performance requirements within allowable tolerances.
Note: Expected artifact for verification are sample executed CPT test procedure and instrument and satellite test plans.

SMBA-198 CPTs shall demonstrate:

a. With the application of known stimuli and appropriate inputs, the test article will produce the expected responses and outputs within allowable tolerances.
b. The operation of redundant circuitry.
c. Satisfactory performance in selected operational modes.
d. When compared to the initial baseline CPT, performance of the test article has been unaffected by the environmental testing program.

SMBA-199 When environmental testing is performed at a given level of assembly, CPTs shall be conducted during the hot and cold extremes of the test for both maximum and minimum input voltages.

SMBA-200 Following integration of the instruments, a CPT will be executed in the configuration expected for environmental testing. Testing shall include operational performance evaluation at minimum and maximum flight voltages.

SMBA-201 All instrument hardware shall be verified by test or inspection for the proper polarity, orientation, and position of all components (sensors, switches, and mechanisms).

SMBA-202 Mechanisms and deployment and latching devices shall be tested to demonstrate adequate functioning both prior to and following exposure to satellite-level environments.

The instrument should be designed to permit full range deployment testing of mechanisms after satellite integration during ground processing at the spacecraft provider facilities.

7.2.3 Limited Performance Tests (LPT) and Aliveness Tests

SMBA-203 Limited Performance Tests (LPTs) shall be conducted at the subsystem, instrument, and satellite levels of assembly, where CPTs are not warranted, to demonstrate that the hardware and software performance has not been degraded by environmental testing/exposure. LPTs are a specific subset of the overall CPT test sequence.
Note: Expected artifact for verification are sample executed LPT test procedure and instrument and satellite test plans.

SMBA-204 LPTs shall demonstrate:

a. With the application of known stimuli and appropriate inputs, the test article will produce the expected responses and outputs within allowable tolerances.
b. The operation of redundant circuitry.
c. Satisfactory performance in selected operational modes.
d. When compared to the initial baseline LPT, performance of the test article has been unaffected by the environmental testing program.

SMBA-205 Electrical Aliveness Tests shall be conducted at the satellite level of assembly, where LPTs and CPTs are not warranted, to verify nominal telemetry response after power on and to demonstrate, to the extent possible by the test configuration, performance has not been degraded by environmental testing/exposure. Electrical Aliveness Tests are a specific subset of the overall CPT/LPT test sequence.
Note: Expected artifact for verification are sample executed Aliveness test procedure and satellite test plan.

7.2.4 NEON Series-1 Ground System Compatibility Tests (GCT)

SMBA-206 At the satellite level, GCTs shall be performed to demonstrate the Mission Operations Center's ground system capability to communicate, command, and control the instrument (up-link and down-link) via satellite mission communications links.

SMBA-207 GCTs shall demonstrate:

a. Simulated normal orbital mission scenarios encompassing launch, systems turn-on, housekeeping, command/control, and stabilization/pointing, including the collecting, processing, and archiving of science data.
b. That the instrument is immune to erroneous commands.
c. Autonomous Safe Mode and simulated anomaly recovery operations.

7.2.5 Flight Hardware Operating Time (Burn-In) and Failure Free Performance

SMBA-208 One thousand (1000) hours of operating/power-on time shall be accumulated on all flight electronic hardware (including all redundant hardware) prior to launch.

SMBA-209 Of the 1000 hours operating/power-on time accumulated on flight hardware prior to launch, at least 200 hours shall be in vacuum split evenly between hot and cold dwells.

SMBA-210 Of the 1000 hours operating/power-on time accumulated on flight hardware prior to launch, the last 350 hours shall be failure-free.

SMBA-211 Ground testing of the integrated instrument flight software with the spacecraft ground system shall accumulate a minimum of 72 hours of error-free, uninterrupted testing using highest fidelity flight software test bed.

7.3 Satellite-Level EMC/EMI Testing

Where not otherwise specified by this document, electromagnetic verification testing should be conducted in accordance with MIL-STD-461G.

SMBA-212 The satellite shall be tested in accordance with Department of Defense Interface Standard - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, MIL-STD-461, Rev G, as tailored by this Section and Section 6.2.

SMBA-213 EMI/EMC tests CE101, CMCE, CE102, CS101, CS114 (CM), CS114 (DM), RE102 (Narrow Band), and RS103 shall be conducted for all electrical flight hardware, as applicable based on test specifications, as shown in Table 7.3-1.

SMBA-214 EMI/EMC tests CS06, CS115, CS116, RE101, RE102 (General), and RS101 shall be conducted for all first unit, new, and re-designed units, as applicable based on test specifications herein. See Table 7.3-1.

Table 7.3-1. Instrument and Subsystem Level Test Applicability Matrix (TBR)

EMI/EMC Test Method	Test Method Reference	Flight Unit Applicability[1]	Test Description	Test Limit or Level	Test Tailoring, Deviations, and Notes
CS101	MIL-STD-461G	ALL	CS, power leads, 30 Hz to 150 kHz	Figures 7.3.4-1 & 7.3.4-2	Section 7.3.4.1
CS06	GSFC-STD-7000B & MIL-STD-462/461C	1st Unit	CS, power leads, transients	GSFC-STD-7000B Figure 2.5-18	Section 7.3.4.2
CS114(CM)	MIL-STD-461G	ALL	CS, bulk cable injection, 10 kHz to 200 MHz	MIL-STD-461G FIGURE CS114-1, Curve 3	Section 7.3.4.3

CS114(DM)	GSFC-STD7000B	ALL	CS, power lead injection, 150 kHz to 50 MHz	Figure 7.3.4-3	Section 7.3.4.4
CS115	MIL-STD-461G	1st Unit	CS, Bulk Cable injection, Impulse excitation (cables only, not leads)	Power Cables: MIL-STD-461G FIGURE CS115-1, 5A max Signal Cables: 1A max	Section 7.3.4.5
CS116	MIL-STD-461G	1st Unit	CS, damped sinusoidal transients, power leads, 10 kHz to 100 MHz (cables only, not leads)	Power Cables: MIL-STD-461G FIGURE CS116-1&2, 5A max Signal Cables: 1A max	Section 7.3.4.6
CE101	GSFC-STD-7000B & MIL-STD-461G	ALL	CE, Power Leads, 30 Hz to 50 MHz	GSFC-STD-7000B Figure 2.5-6	Section 7.3.3.1
CMCE	GSFC-STD-7000B	ALL	Common Mode Conducted Emissions, 10 kHz to 200 MHz	Figure 7.3.3-1	Section 7.3.3.2
CE102	MIL-STD-461G	ALL	CE, Power Leads, 10 kHz to 10 MHz	MIL-STD-461G FIGURE CE102-1, BASIC Curve	Section 7.3.3.3
RS101	MIL-STD-461G	1st Unit	RS, Magnetic Field, 30 Hz to 100 kHz	MIL-STD-461G FIGURE RS101-1	Section 7.3.2.3
RS103	MIL-STD-461G	ALL	RS, Electrical Field, 10 kHz to 40 GHz	See Section 6.2	Section 7.3.2
RE101	MIL-STD-461G	1st Unit	RE, Magnetic Field, 30 Hz to 100 kHz	MIL-STD-461G FIGURE RE101-2	Section 7.3.1.3
RE102	MIL-STD-461G	ALL	RE, Electrical Field, 2 MHz to 18 GHz	Tables 7.3.1-3	Section 7.3.1

Notes:
[1] 1st Unit also refers to new and re-designed units per SMBA-214.

SMBA-215 For all MIL-STD-461G EMI/EMC testing, the line impedance stabilization network (LISN) in MIL-STD-461G shall be tailored, as necessary, to be more representative of the spacecraft power bus impedance. When not otherwise specified, the 5 microhenry LISN specified in Section A.4.3.6, Figures A-2 & A-3 of MIL-STD-461G will be utilized in place of the standard 50 microhenry LISN.

Flight-like cables (constructed with expected flight wire groupings, twisting, shielding, and shield terminations) should be used for all EMC testing.

All connectors that are not used for flight should have flight-quality EMI-preventative covers that are captive when installed for launch. For system-level testing, connectors that are not used for flight will be capped with EMI-preventative covers.

7.3.1 Radiated Emissions (RE)

The instrument will meet the radiated emissions limits specified in the following subsections.

7.3.1.1 RE, Electric Field, Standard Bandwidth(s)

The purpose of this requirement is to verify that electric field emissions from the satellite and its associated cabling do not interfere with on-board RF receivers or launch vehicle receivers. The specific requirements for the test, including test methods, are detailed in requirements below:

The basic test method of MIL-STD-461G RE102 will apply for the Satellite Radiated (E-Field, Standard Bandwidth) Emissions Test.

SMBA-216	The Satellite RE102 (standard) test shall be conducted in the Science Operations Mode.

SMBA-217	The RE102 EMI antennas shall be positioned such that emissions will be positioned at critical instrument locations as specified in the ICD.

SMBA-218	The frequency range for the RE102 test shall be 2 MHz to 18 GHz for the Nominal Operational Mode and 14 kHz to 18 GHz for the Launch Mode.

SMBA-219	The RE102 standard bandwidth limits shall be as specified in Table 7.3.1-1.

Table 7.3.1-1. Unintentional Radiated Electric Field Emissions (General), 14kHz to 18 GHz, Standard MIL-STD-461G Bandwidths

Frequency (MHz)	Limit Level (dBμV/m)	6dB Resolution Bandwidth (Hz)	Receiver Protected
0.014[1]	58	1K	
0.018[1]	56	1K	
0.15[1]	56	1K	
0.15[1]	56	10K	
2	56	10K	
30	56	10K	
30	56	100K	
1000	56	100K	
1000	56	1M	
1127	56	1M	
1127	42	1M	GPS-L2
1327	42	1M	GPS-L2
1327	56	1M	
1475	56	1M	
1475	43	1M	GPS-L1
1675	43	1M	GPS-L1
1675	56	1M	
2000	56	1M	
2000	45	1M	SC-CMD
2150	45	1M	SC-CMD
2150	56	1M	
4000	56	1M	
4000-18000	+20 dB per decade	1M	
18000	69	1M	

Notes:
[1] Applies only to Launch Mode

7.3.1.2 RE, Electric Field, Narrow Bandwidth(s)

The purpose of this requirement is to verify that electric field emissions from the satellite and its associated cabling do not interfere with on-board RF receivers or launch vehicle receivers. The specific requirements for the test, including test methods, are detailed in requirements below:

The basic test method of MIL-STD-461G RE102 applies for the Satellite Radiated (E-Field, Narrow Bandwidth) Emissions Test.

SMBA-220 The radiated emission measurement bandwidths shall be reduced from the MIL-STD-461G values as specified in Table 7.3.1-2 to show compliance with the receiver notches. Only the selected frequency ranges in Table 7.3.1-2 are applicable to the Radiated E-Field, Narrow Bandwidth, Emissions Test.

SMBA-221 The Satellite RE102 (Narrowband) test shall be conducted in the Science Operations Mode.

SMBA-222 The RE102 narrow bandwidth limits shall be as specified in Table 7.3.1-2.

SMBA-223 Unintentional radiated emission measurements in the specified Table 7.3.1-2 receiver bands shall be made in accordance with MIL-STD-461G RE102 with the EMI receiver preceded by a low-noise preamplifier at the measurement antenna such that the test system noise figure is less than or equal to 3 dB.

Table 7.3.1-2. Unintentional Radiated Electric Field Emissions, Receiver Frequencies, Narrowband MIL-STD-461G Scans (Modified Bandwidths)

Frequency (MHz)	Limit Level (dBμV/m)	6dB Resolution Bandwidth (Hz)	Receiver Protected
1127	42		
1127	39	100K	
1208	39	100K	
1208	12[1]	1000[1]	GPS-L2
1244	12[1]	1000[1]	GPS-L2
1244	39	100K	
1327	39	100K	
1327	42		
1475	43		
1475	39	100K	
1556	39	100K	
1556	14	1000	GPS-L1
1592	14	1000	GPS-L1
1592	39	100K	
1675	39	100K	
1675	43		
2000	45	1M	
2025	45	1M	
2025	15	1000	SC-CMD
2110	15	1000	SC-CMD
2110	45	1M	
2150	45	1M	

Notes:
[1] Applies only if GPS-L2 is implemented on the satellite, otherwise limit is 39 dBuV/m at 100K RBW

7.3.1.3 RE, Magnetic Field, RE101

SMBA-224 Radiated AC magnetic field emissions from the instrument shall not exceed the levels shown in Figure RE101-2 at a distance of 7 cm per MIL-STD-461G, RE101 from 30 Hz to 100 kHz.
Note: Test requirement applies to new or redesigned hardware; otherwise previous test artifacts may be substituted in lieu of testing.

7.3.1.4 RE, SMBA Instrument Compatibility

SMBA-225 The instrument provider shall confirm performance and evaluate the ability to meet the radiated emissions requirements shown in Tables 7.3.1-3 and 7.3.1-4 at the instrument apertures.

Spacecraft components and other instruments will meet the radiated emissions requirements shown in Table 7.3.1-3. The SMBA instrument will not be susceptible below these levels at angles > TBD degrees from the aperture boresight(s).

Table 7.3.1-3. Intentional and Unintentional Radiated Electric Field Emissions Limits at the SMBA Instrument Interface, 18 to 235 GHz (TBR)

Frequency Range From (MHz)	To (MHz)	Emission Limits (dBµV/m) [1]	Bandwidth	Susceptible Receiver or Intentional Transmitter
18000	31800	69 - 74	1 MHz	General Requirement
21200	21400	TBR	TBR	SMBA K-Band
22210	22500	TBR	TBR	SMBA K-Band
23567.25	24032.75	45	100 Hz	Heritage Channel
25500	27000	154	300 MHz	SC Ka Transmitter
31241.5	31558.5	52	1000 Hz	Heritage Channel
31558.5	31800	TBR	TBR	SMBA Ka Band
31800	40000	74 - 76	1 MHz	General Requirement
50141.5	50458.5	56	180 MHz	Heritage Channel
50200	50400	TBR	TBR	SMBA V-Band
51425	52095	61	400 MHz	Heritage Channel
52465	53135	61	400 MHz	Heritage Channel
53316	53596	60	170 MHz	Heritage Channel
53596	53876	60	170 MHz	Heritage Channel
54065	55782.25	60	400 MHz	Heritage Channel
56889.22	57691.46	47	330 MHz	Heritage Channel
52600	59300	TBR	TBR	SMBA V-Band
86000	92000	TBR	TBR	SMBA W-Band
86500	89900	71	2000 MHz	Heritage Channel
100000	102000	TBR	TBR	TBR
105000	110000	TBR	TBR	TBR
110000	122250	TBR	TBR	SMBA F-Band
148000	151000	TBR	TBR	SMBA D-Band
162950	168050	84	3000 MHz	Heritage Channel
164000	167000	TBR	TBR	SMBA D-Band
174610	178010	86	2000 MHz	Heritage Channel
174800	191800	TBR	TBR	SMBA G-Band
177110	180510	87	2000 MHz	Heritage Channel

179460	181160	84	1000 MHz	Heritage Channel
180660	182360	84	1000 MHz	Heritage Channel
181885	182735	81	500 MHz	Heritage Channel
183885	184735	81	500 MHz	Heritage Channel
184260	185960	84	1000 MHz	Heritage Channel
185460	187160	84	1000 MHz	Heritage Channel
186110	189510	87	2000 MHz	Heritage Channel
188610	192010	87	2000 MHz	Heritage Channel
200000	209000	TBR	TBR	SMBA J-Band
217000	231500	TBR	TBR	SMBA J-Band

Notes:

[1] General requirement is 69 dBµV/m at 18 GHz and increases +20 dB/decade to 31.8 GHz, then 74 dBµV/m increasing +20 dB/decade. Limits apply at the respective channel Bandwidths at the location of the SMBA apertures. Levels above the general requirement are intentional spacecraft emission limits. This table assumes the emitter is located at an angle outside the critical stayout zones identified in the ICD. Less stringent limits may apply at angles farther from these areas. Refer to Table 7.3.1-4. No RF energy may be directed to within critical stay out zones referenced from the SMBA scan planes.

Table 7.3.1-4. SMBA Receive Sensitivity (TBR)

(This table will be revised to reflect the specifics of the instrument chosen)

Frequency Band	Center Frequency	Max. Bandwidth	Frequency Range		EMI Sensitivity of SMBA Passbands [4]			
	(MHz)	(MHz)	From (MHz)	To (MHz)	dBm	dBµV/m Boresight ≤15.5° [1][5]	dBµV/m <40° [2] [5]	dBµV/m ≥40° [3] [5]
K	TBR	TBR	20000	25000	TBR	TBR	TBR	TBR
K	23800	270	21300	22000	-60	72.79	115.51	129.01
			22000	22800	-69	63.79	106.51	120.01
			22800	23567.25	-91	42.22	84.94	98.44
			23567.25	24032.75	-131	2.45	45.17	58.67
			24032.75	24800	-91	42.67	85.39	98.89
			24800	25900	-70	64.07	106.79	120.29
			25900	26300	-27	107.07	149.79	163.29
			26300	26550	-17	117	160	173
			26550	26850	-17	117	160	173
			26850	28900	-17	118	160	174
Ka	TBR	TBR	31000	33000	TBR	TBR	TBR	TBR
Ka	31400	180	28900	30400	-53	82.76	132.17	161.17
			30400	31241.5	-93	43.1	92.51	121.51
			31241.5	31558.5	-133	3.26	52.67	81.67
			31558.5	32400	-93	43.42	92.83	121.83
			32400	33900	-54	82.73	132.14	161.14
V	TBR	TBR	50000	62000	TBR	TBR	TBR	TBR
V	50300	180	47641.5	49300	-53	79.53	136.58	149.58
			49300	50141.5	-93	39.75	96.8	109.8
			50141.5	50458.5	-133	-0.15	56.9	69.9
			50458.5	51300	-93	39.95	97	110
			51300	52800	-53	80.15	137.2	150.2
V	51760	400	49260	50760	-49	83.98	141.03	154.03
			50760	51425	-89	44.16	101.21	114.21
			51425	52095	-129	4.27	61.32	74.32
			52095	52760	-89	44.38	101.43	114.43
			52760	54260	-49	84.56	141.61	154.61
V	52800	400	50300	51800	-49	83.98	141.03	154.03
			51800	52465	-89	44.16	101.21	114.21
			52465	53135	-129	4.27	61.32	74.32
			53135	53800	-89	44.38	101.43	114.43
			53800	55300	-49	84.56	141.61	154.61
V	53596 ± 115	170	51096	52596	-50	83.11	140.16	153.16
			52596	53336	-90	43.3	100.35	113.35
			53336	53596	-130	3.45	60.5	73.5
			53596	53856	-130	3.44	60.49	73.49

			53856	54596	-90	43.51	100.56	113.56
			54596	56096	-50	83.68	140.73	153.73
V	54400	400	51900	53400	-49	84.25	141.3	154.3
			53400	54065	-89	44.42	101.47	114.47
			54065	54735	-129	4.53	61.58	74.58
			54735	55400	-89	44.64	101.69	114.69
			55400	56900	-49	84.81	141.86	154.86
V	54940	400	52440	53940	-49	84.34	141.39	154.39
			53940	54605	-89	44.51	101.56	114.56
			54605	55275	-129	4.62	61.67	74.67
			55275	55940	-89	44.72	101.77	114.77
			55940	57440	-49	84.89	141.94	154.94
V	55500	330	53000	54500	-50	83.43	140.48	153.48
			54500	55217.75	-90	43.6	100.65	113.65
			55217.75	55782.25	-130	3.71	60.76	73.76
			55782.25	56500	-90	43.81	100.86	113.86
			56500	58000	-50	83.98	141.03	154.03
V	57290.344	330	54790.34	56290.34	-50	83.71	140.76	153.76
			56290.34	57017.59	-90	43.88	100.93	113.93
			57017.59	57563.09	-130	3.98	61.03	74.03
			57563.09	58290.34	-90	44.08	101.13	114.13
			58290.34	59790.34	-50	84.24	141.29	154.29
V	57290.344 ± 217	78	54790.34	56290.34	-53	80.71	137.76	150.76
			56290.34	57008.4	-93	40.88	97.93	110.93
			57008.4	57138.19	-133	0.95	58	71
			57138.19	57442.49	-93	40.98	98.03	111.03
			57442.49	57572.19	-133	1.01	58.06	71.06
			57572.19	58290.34	-93	41.08	98.13	111.13
			58290.34	59790.34	-53	81.24	138.29	151.29
V	57290.344 ± 322.2 ± 48	36	54790.34	56290.34	-53	80.71	137.76	150.76
			56290.34	56889.22	-93	40.87	97.92	110.92
			56889.22	56951.02	-133	0.93	57.98	70.98
			56951.02	56985.22	-93	40.93	97.98	110.98
			56985.22	57047.02	-133	0.94	57.99	70.99
			57047.02	57533.66	-93	40.98	98.03	111.03
			57533.66	57595.46	-133	1.02	58.07	71.07
			57595.46	57629.66	-93	41.03	98.08	111.08
			57629.66	57691.46	-133	1.04	58.09	71.09
			57691.46	58290.34	-93	41.09	98.14	111.14
			58290.34	59790.34	-53	81.24	138.29	151.29
V	57290.344 ± 322.2 ± 22	16	54790.34	56290.34	-57	76.71	133.76	146.76
			56290.34	56931.72	-97	36.88	93.93	106.93
			56931.72	56960.52	-137	-3.07	53.98	66.98
			56960.52	56975.72	-97	36.93	93.98	106.98
			56975.72	57004.52	-137	-3.06	53.99	66.99
			57004.52	57576.16	-97	36.98	94.03	107.03
			57576.16	57604.96	-137	-2.97	54.08	67.08
			57604.96	57620.16	-97	37.03	94.08	107.08
			57620.16	57648.96	-137	-2.97	54.08	67.08
			57648.96	58290.34	-97	37.08	94.13	107.13
			58290.34	59790.34	-57	77.24	134.29	147.29
V	57290.344 ± 322.2 ± 10	8	54790.34	56290.34	-60	73.71	130.76	143.76
			56290.34	56951.02	-100	33.88	90.93	103.93
			56951.02	56965.22	-140	-6.07	50.98	63.98
			56965.22	56971.02	-100	33.93	90.98	103.98
			56971.02	56985.22	-140	-6.07	50.98	63.98
			56985.22	57595.46	-100	33.98	91.03	104.03
			57595.46	57609.66	-140	-5.97	51.08	64.08
			57609.66	57615.46	-100	34.03	91.08	104.08
			57615.46	57629.66	-140	-5.97	51.08	64.08
			57629.66	58290.34	-100	34.08	91.13	104.13
			58290.34	59790.34	-60	74.24	131.29	144.29
V	57290.344 ± 322.2 ± 4.5	3	54790.34	56290.34	-64	69.71	126.76	139.76
			56290.34	56960.65	-104	29.88	86.93	

			56960.65	56966.6	-144	-10.07	46.98	59.98
			56966.6	56969.65	-104	29.93	86.98	99.98
			56969.65	56975.6	-144	-10.07	46.98	59.98
			56975.6	57605.09	-104	29.98	87.03	100.03
			57605.09	57611.04	-144	-9.97	47.08	60.08
			57611.04	57614.09	-104	30.03	87.08	100.08
			57614.09	57620.04	-144	-9.97	47.08	60.08
			57620.04	58290.34	-104	30.08	87.13	100.13
			58290.34	59790.34	-64	70.24	127.29	140.29
W	TBR	TBR	85000	92000	TBR	TBR	TBR	TBR
W	88200	2000	83250	84750	-42	94.93	151.47	163.47
			84750	86500	-82	55.04	111.58	123.58
			86500	89900	-122	15.35	71.89	83.89
			89900	91650	-82	55.65	112.19	124.19
			91650	93150	-42	95.75	152.29	164.29
F	TBD	TBD	108000	123000	TBD	TBD	TBD	TBD
D	TBR	TBR	164000	167000	TBR	TBR	TBR	TBR
D	165500	3000	158000	160475	-40	97.42	164.63	176.63
			160475	162950	-80	57.48	124.69	136.69
			162950	168050	-120	17.6	84.81	96.81
			168050	170525	-80	57.71	124.92	136.92
			170525	173000	-40	97.76	164.97	176.97
G	TBR	TBR	173000	193000	TBR	TBR	TBR	TBR
G	183310±7000	2000	173810	174210	-39	98.46	166.86	179.36
			174210	174610	-79	58.51	126.91	139.41
			174610	178010	-119	18.53	86.93	99.43
			178010	188610	-79	58.88	127.28	139.78
			188610	192010	-119	19.22	87.62	100.12
			192010	192410	-79	59.24	127.64	140.14
			192410	192810	-39	99.29	167.69	180.19
G	183310±4500	2000	176310	176710	-39	98.58	166.98	179.48
			176710	177110	-79	58.64	127.04	139.54
			177110	180510	-119	18.65	87.05	99.55
			180510	186110	-79	58.88	127.28	139.78
			186110	189510	-119	19.11	87.51	100.01
			189510	189910	-79	59.12	127.52	140.02
			189910	190310	-39	99.17	167.57	180.07
G	183310±3000	1000	177810	178635	-42	95.66	164.06	
			178635	179460	-82	55.71	124.11	136.61
			179460	181160	-122	15.73	84.13	96.63
			181160	185460	-82	55.88	124.28	136.78
			185460	187160	-122	16.04	84.44	96.94
			187160	187985	-82	56.06	124.46	136.96
			187985	188810	-42	96.11	164.51	177.01
G	183310±1800	1000	179010	179835	-42	95.71	164.11	176.61
			179835	180660	-82	55.77	124.17	136.67
			180660	182360	-122	15.78	84.18	96.68
			182360	184260	-82	55.88	124.28	136.78
			184260	185960	-122	15.98	84.38	96.88
			185960	186785	-82	56	124.4	136.9
			186785	187610	-42	96.05	164.45	176.95
G	183310±1000	500	179810	180847.5	-45	92.75	161.15	173.65
			180847.5	181885	-85	52.8	121.2	133.7
			181885	182735	-125	12.82	81.22	93.72
			182735	183885	-85	52.88	121.28	133.78
			183885	184735	-125	12.95	81.35	93.85
			184735	185772.5	-85	52.96	121.36	133.86
			185772.5	186810	-45	93.01	161.41	173.91
J	TBD	TBD	200000	235000	TBD	TBD	TBD	TBD

Notes:

[1] For susceptibility to a source within 15.5° (TBS) of the SMBA antenna boresight, the "Boresight ≤15.5°" column should be used for worst-case analysis.

[2] For susceptibility to a source between 15.5° (TBS) and 40° (TBS) of the SMBA antenna boresight, the "<40°" column should be used for worst-case analysis.

[3] For susceptibility to a source at a 40°(TBS) or greater angle from the SMBA antenna boresight, the "≥40°" column should be used for worst-case analysis.

[4] Limits apply at the Max Bandwidth of the respective SMBA channel.

[5] See Spacecraft-to-SMBA Instrument MICD (NS1-SMBA-xxx-xxxx) for EMI Boresight Stayout Zones.

7.3.2 Radiated Susceptibility (RS)

SMBA-226 Instrument systems shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated in the satellite, individual equipment, or subsystem specification, when subjected to the radiated susceptibility tests defined in the following subsections.

SMBA-227 For all RS Electric Field Tests, sufficient filtering shall be employed on EMI test equipment to prevent broadband noise and harmonics from interfering with the SMBA receiver bands (see Table 7.3.1-4, SMBA Receive Sensitivity).

SMBA-228 For all Radiated Susceptibility Electric Field Tests, analysis shall show that the output power applied at the test antenna produces the required levels at the Instrument and ensures that measured levels are not the result of peak resonances or reflections.

7.3.2.1 RS, Electric Field, Orbit and Launch Levels

The purpose of this requirement is to verify the ability of the instruments, spacecraft hardware, and associated cabling to withstand electric fields generated from intentional RF transmitters during all phases of the project, including on-orbit, on the launch vehicle, at the launch site, and all test environments. The specific requirements for the test, including test methods, are detailed in requirements below:

SMBA-229 For the Satellite Orbit and Launch Mode Radiated (E-Field) Susceptibility Tests, the MIL-STD-461G RS103 test method shall apply.

SMBA-230 The frequency range for the RS103 tests shall be 14 kHz to 18 GHz for the Launch Mode and 2 MHz to 40 GHz for the on-orbit Operational Mode. Note: Testing above 18 GHz is limited to frequencies identified in Table 6.2-1.

SMBA-231 The dwell time for each frequency step shall be tailored from the MIL-STD-461G default of 3 seconds to one scan period or integer multiple of the scan period at the instrument locations.

SMBA-232 For the Satellite RS103 Test (on-orbit configuration), the levels in Table 6.2-1 (TBD) (Columns 3 and 4) shall apply.

The limits were derived from data and analyses compiled by the Department of Defense Joint Spectrum Center (JSC-CR-06-070, Space Vehicle RF Environments) and were scaled for an ~824 km orbit.

SMBA-233 For the On-Orbit External levels for frequency ranges 200 MHz to 18 GHz (as identified in Table 6.2-1), at least one test antenna position shall radiate within the Field of View (aperture) of the Instrument.

SMBA-234 For the Satellite RS103 Test (Launch Configuration), the limits in Table 6.2-1 (TBD) (Columns 1 and 2) shall apply.

7.3.2.2 RS, Electric Field, Satellite Transmitter Frequencies

The purpose of this requirement is to verify the ability of the satellite to withstand electric fields generated from intentional RF transmitters during the on-orbit phase of the mission. The specific requirements for the test, including test methods, are detailed in requirements below:

SMBA-235 The MIL-STD-461G RS103 test method shall apply for the satellite transmitter frequency radiated susceptibility test with the exceptions in SMBA-236 through SMBA-239.

SMBA-236 The susceptibility scans should be performed using the modulation format of the specific transmitter(s) that will be producing the fields on the satellite; however, if this is not known or test equipment limitations prevent such modulation, then modulation shall be pulse modulated (on/off ratio of 40 dB minimum) at a 1 kHz rate with a 50% duty cycle per MIL-STD-461G.

SMBA-237 Minimum scan rates and step sizes shall be reduced from the MIL-STD-461G per Table 7.3.2-1.

SMBA-238 Minimum dwell times for stepped scans shall be tailored from the MIL-STD-461G requirement of 3 seconds per Table 7.3.2-1 to one scan period or integer multiple of the scan period.

SMBA-239 Satellite transmitter frequency RS103 intentional test limits and exposure levels for the instrument shall be as specified in Table 7.3.2-1.

Table 7.3.2-1. Spacecraft Transmitter Frequency Susceptibility Test Levels (TBS/TBR)

Transmitter Name	Start Frequency (MHz)	Stop Frequency (MHz)	Analog Scan, Maximum Rate (MHz/sec)	Stepped Scan, Maximum Step Size (*f0)	Minimum Dwell Time per step (Sec.)	Level (V/m) [1]
S-Band (Center Freq)	TBS	TBS	NA	NA	64	36
S-Band (Full BW)	TBR 2200	TBR 2290	0.0166	0.00001 (1.0E-05)	3 [2]	36
X-Band, (Center)	TBS	TBS	NA	NA	64	45
X-Band, (Full BW)	TBR 7750	TBR 7900	0.166	0.00006 (6.0E-05)	3 [2]	45
Ka-Band, (Center)	TBS	TBS	NA	NA	64	50, 1.0 [3]

Ka-Band, (Full BW)	TBR 25500	TBR 27000	2.0	0.0002 (2.0E-04)	3 [2]	50, 1.0 [3]

Notes:
[1] The limits above are generic levels and include a ~6dB margin. These levels may be tailored to expected distances from the specified emitting antenna, however margin must be maintained to account for multipath effects.
[2] MIL-STD-461G default is 3 seconds, Exceptions – test should use SMBA scan period(s) (see SMBA Instrument to Spacecraft ICD (NS1-SMBA-xxx-xxxx)).
[3] Reduced level within critical stayout zone of SMBA boresight.

7.3.2.3 RS, Magnetic Field, RS101

SMBA-240 The instrument shall operate within specified performance, when subjected to the magnetic field levels shown in MIL-STD-461G, Figure RS101-1 (Navy parameters) from 30 Hz to 100 kHz per the RS101 test method.
Note: Test requirement applies to new or redesigned hardware; otherwise previous test artifacts may be substituted in lieu of testing.

7.3.3 Conducted Emissions Tests

7.3.3.1 CE101

SMBA-241 The instrument shall limit reflected ripple to the CE101 differential mode current level shown in GSFC-STD-7000B, Section 2.5.2.1.1, Figure 2.5-6 when tested from 30 Hz to 50 MHz in a true differential mode per the test method.
Note 1: The CE101 differential mode low-frequency plateau limit from 30 Hz to 2 kHz is determined by adding a factor of 20*log (load current in Amperes) to a base of 106 dBμA.
Note 2: The capacitor network specified by GSFC-STD-7000B should be inserted on the power supply side of the LISN specified in SMBA-215.

7.3.3.2 CMCE

SMBA-242 A Common Mode Conducted Emissions (CMCE) Test shall be performed from 10 kHz to 200 MHz on the instrument power and signal cables per GSFC-STD-7000B Section 2.5.2.1.2 except that the 5 microHenry LISN specified SMBA-215 may be used with the capacitor network on the power supply side of the LISN.

SMBA-243 Common Mode Conducted Emissions (CMCE) from Instrument power and signal cables shall not exceed the limits specified in Figure 7.3.3-1.

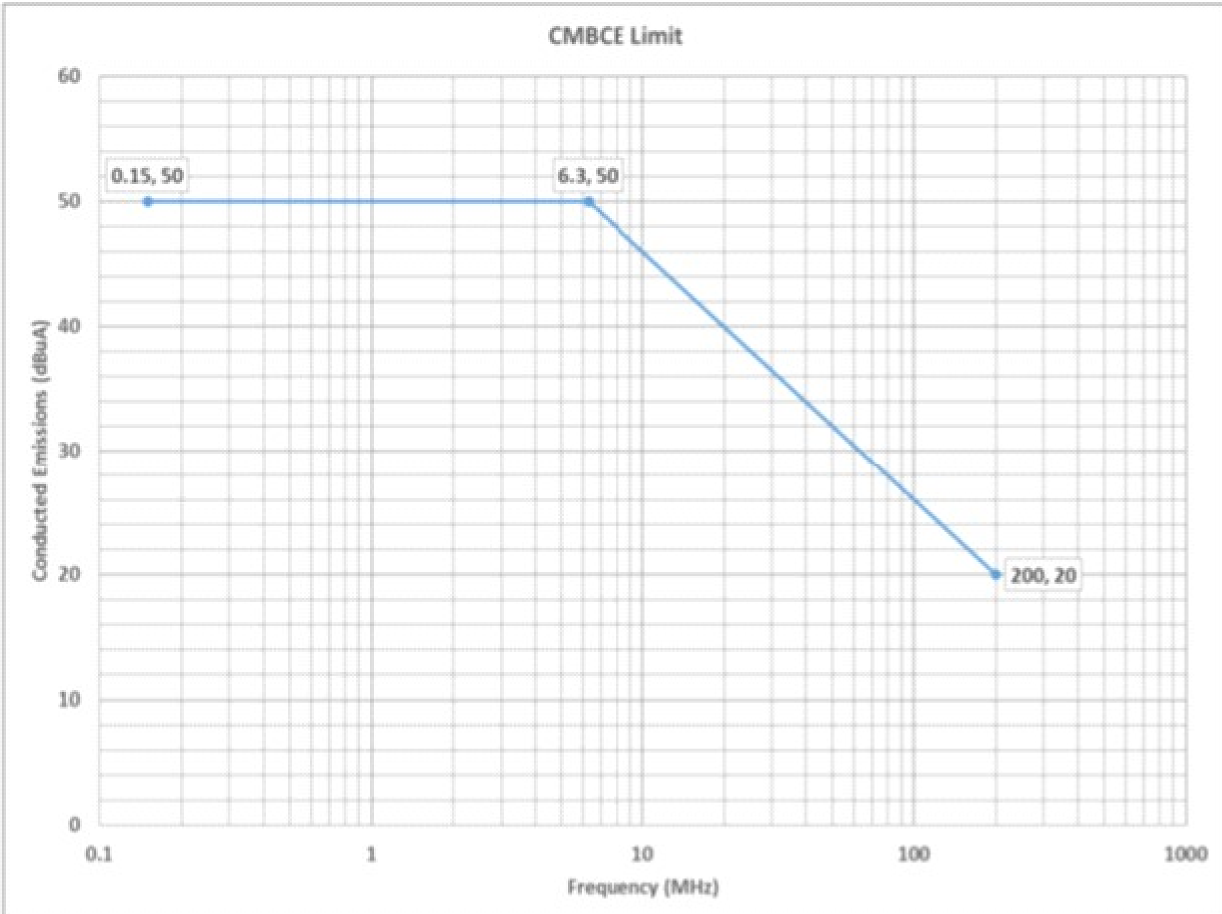

Figure 7.3.3-1. Common Mode Conducted Emissions (CMCE) Limit [TBS]

7.3.3.3 CE 102

SMBA-244	The instrument shall limit reflected ripple from 10 kHz to 10 MHz to the CE102 voltage level shown in MIL-STD-461G, Figure CE102-1, when tested per the CE102 Test method using LISNs specified in SMBA-215.

7.3.4 Conducted Susceptibility Tests

7.3.4.1 CS101

SMBA-245	The instrument shall operate within specified performance, when power leads are subjected to the voltage levels shown in Figure 7.3.4-1 from 30 Hz to 150 kHz per the MIL-STD-461G CS101 test method. The requirement is also met when the power source is adjusted to dissipate the power level shown in Figure 7.3.4-2 in a 0.5 ohm load and the instrument is not susceptible.

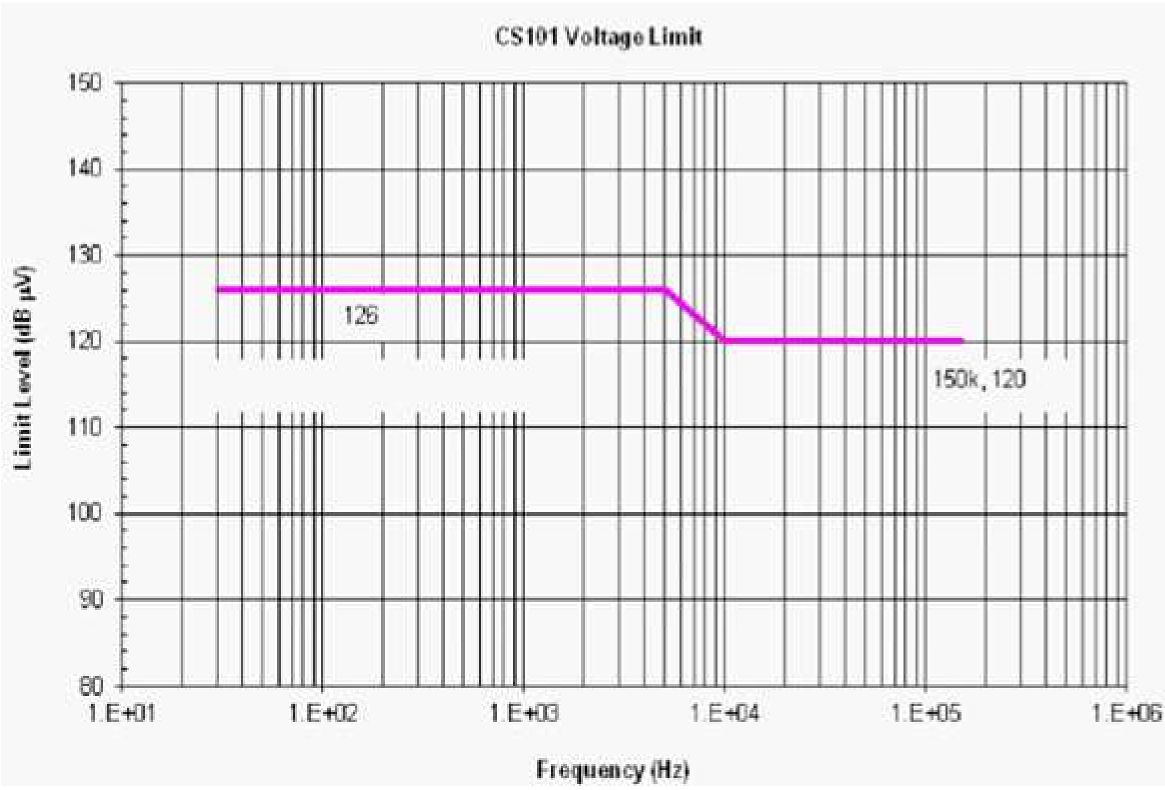

Figure 7.3.4-1. Instrument Power Lead Conducted Susceptibility (CS101) Voltage Limit

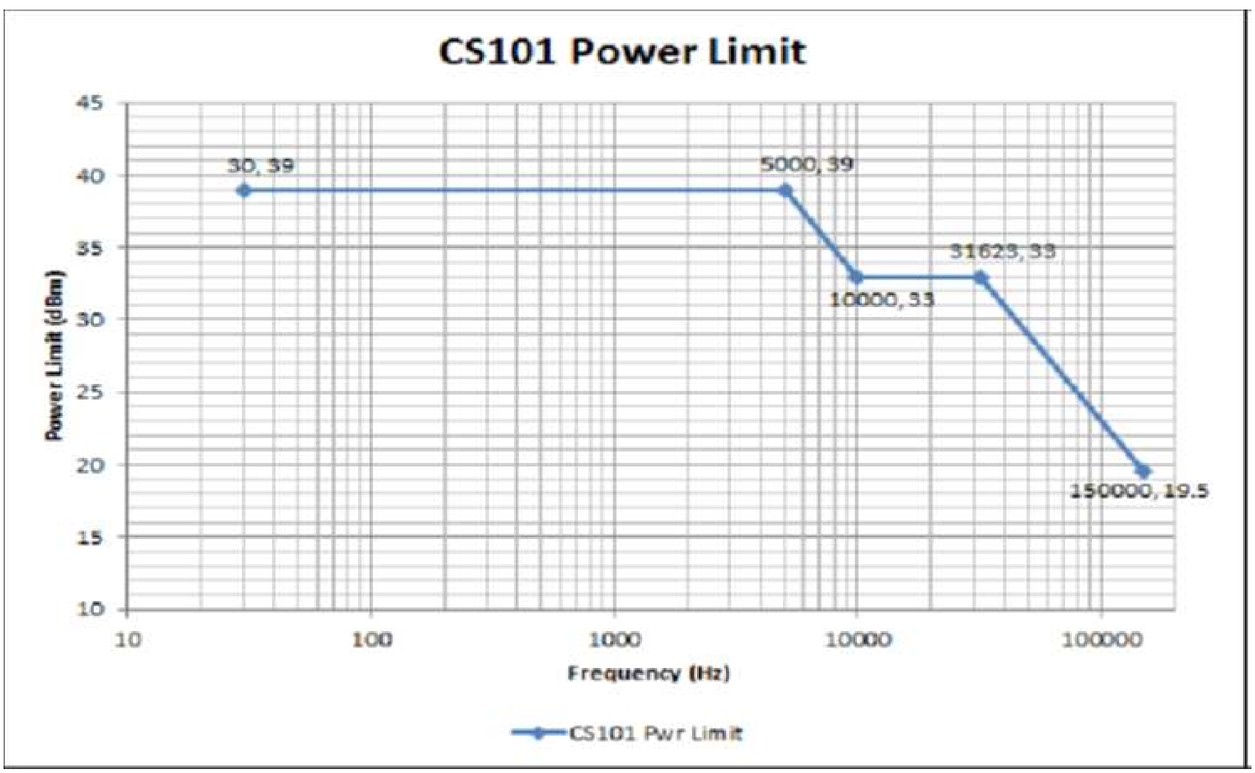

Figure 7.3.4-2. Instrument Power Lead Conducted Susceptibility (CS101) Power Limit

7.3.4.2 CS06

SMBA-246	The instrument shall operate without malfunction, loss of capability, change of operation state/mode, memory changes or need for outside intervention when subjected to the MIL-STD-462 test CS06 (negative transient only) per GSFC-STD-7000B section 2.5.2.2.3, with a negative spike tolerance of +5.0V/-0V.

For the (full level) CS06 test, each spike will be superimposed on the powerline voltage (+30 V) having the waveform shown in GSFC-STD-7000B, Section 2.5.2.2.3, Figure 2.5-18, and applied to the power input leads once per second, for a total test period of 1 minute in duration. The in-line 5 microHenry LISNs specified in SMBA-215 along with any necessary in line inductance in series or capacitance across the LISN outputs may be used in place of the GSFC-STD-7000B specified capacitor network.

7.3.4.3 CS114CM

SMBA-247	The Instrument shall operate within specified performance, when spacecraft interface power and signal cables are subjected to the calibrated limits shown in Curve 3 of MIL-STD-461G Figure CS114-1 from 10 kHz to 200 MHz per the CS114 test method (Common-Mode).

7.3.4.4 CS114DM

SMBA-248	The Instrument shall operate within specified performance, when the high-side power leads are subjected to an injection probe drive level which has been precalibrated to the level shown in Figure 7.3.4-3 from 150 kHz to 50 MHz. The test method is per MIL-STD-461G CS114 (high power leads with returns and grounds removed) except that a capacitor (10uF minimum) is placed across the supply and return terminals of the LISN outputs and the current limit specified in Figure 7.3.4-3 is used.

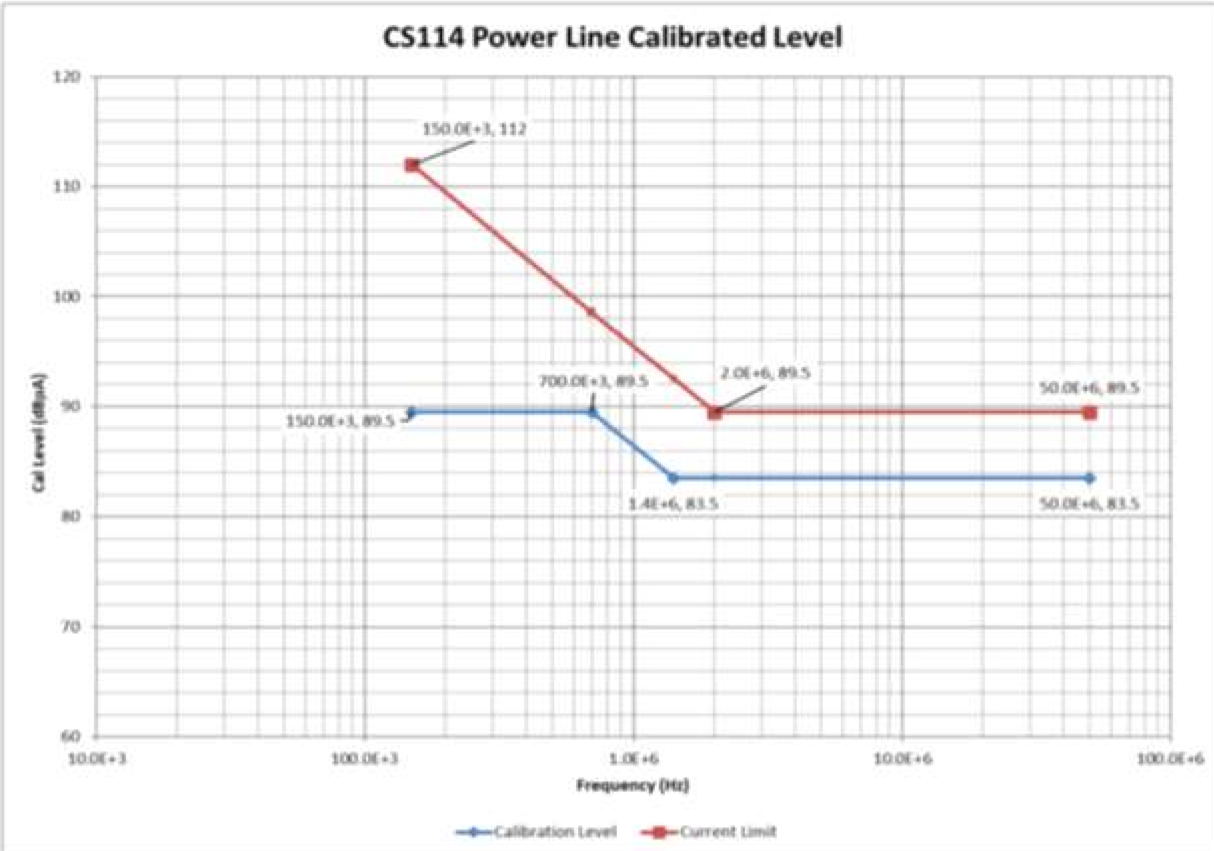

Figure 7.3.4-3. Conducted Susceptibility, Power Line Injection (CS114), Calibrated Level and Current Limit

7.3.4.5 CS115

SMBA-249	The instrument shall operate without malfunction, loss of capability, change of operation state/mode, memory changes or need for outside intervention when spacecraft interface power cables are subjected to a pre-calibrated signal having rise and fall times, pulse width, and amplitude as specified in MIL-STD-461G Figure CS115-1 at a 30 Hz rate for one minute per the CS115 test method. Signal cables are to be subjected to the same test except the amplitude may be reduced to 1A max. Note: Only Bulk Cable Common Mode Injection is applicable.

7.3.4.6 CS116

SMBA-250	The instrument shall operate without malfunction, loss of capability, change of operation state/mode, memory changes or need for outside intervention when spacecraft interface power cables are subjected to a signal having the waveform shown in MIL-STD-461G Figure CS116-1 and the amplitude for the curve shown in Figure CS116-2 except the Peak current is 5 Amps using the CS116 test method. Signal cables are to be subjected to the same test except the Peak current may be reduced to 1A max. Note: Only Bulk Cable Common Mode Injection is applicable.

7.4 Structural and Mechanical Verification Requirements

The instrument will experience no unintentional change of state when subjected to the specified dynamic environments specified herein.

SMBA-251	Structural and mechanical tests and analyses defined in Table 7.4-1 shall be conducted to demonstrate that the flight hardware is qualified for the expected mission environments and that the design of the hardware complies with the specified verification requirements such as factors of safety, interface compatibility, structural reliability, workmanship, and associated elements of system safety.

Table 7.4-1. Structural and Mechanical Verification Test Requirements

Requirement	Payload/ Spacecraft	Subsystem/ Instrument	Unit (Component) Including Instrument Units (Components)
Structural Loads			
Modal Survey	*	T^2	*
Design Qualification	*	A, T/A^1	*
Structural Reliability Primary & Secondary Structure	*	$(A,T)^1$	*
Vibroacoustics			
Acoustics	T	T^2	T^2
Random Vibration	T^2	T^2	T
Sine Vibration	T^3,T^4	T^3	T^3
Mechanical Shock	T	T^5	T^5
Mechanical Function	A, T	A, T	-
Pressure Profile	-	A, T^2	A
Mass Properties	A/T	A, T^2	*

Notes:
* = May be performed at payload or component level of assembly if appropriate.
A = Analysis required.
T = Test required.
A/T = Analysis and/or Test.
$A,T/A^1$ = Analysis and Test or Analysis only, using no-test factors of safety of 2.0 on yield and 2.6 on ultimate (with NASA concurrence).
$(A,T)^1$ = Combination of fracture analysis and proof tests on selected elements, with special attention given to beryllium, composites, bonded joints, and weldments.
T^2 = Test to be performed unless assessment justifies deletion.
T^3 = Test performed to simulate low frequency transient vibration and any sustained periodic mission environment, or to satisfy other requirement such as strength qualification.
T^4 = Test performed for ELV payloads, if practicable, to simulate transient and any sustained periodic vibration mission environment.
T^5 = Test required for self-induced shocks but may be performed at payload level of assembly for externally induced shocks.

7.4.1 Structural Loads Qualification

SMBA-252 The strength qualification test shall be accompanied by a stress analysis that demonstrates a positive margin on ultimate at loads equal to 1.4 times the limit load for metallic structures and 1.6 times limit load for composite, including Beryllium, structures for all ultimate failure modes such as fracture or buckling.

SMBA-253 Acceptance structural loads test profile shall be the limit load for 5 cycles at full level per axis if using the sine burst technique, or a sustained 30 seconds of loading at full level otherwise (centrifuge or static load).

SMBA-254 Qualification/Protoflight structural loads test profile shall be 1.25 times limit load for 5 cycles at full level per axis if using the sine burst technique, or a sustained 30 seconds (Protoflight)/60 seconds (Qualification) of loading at full level otherwise (centrifuge or static load).

SMBA-255	All composite and Beryllium structures and structural bonded joints cannot be qualified by analysis alone; they shall be proof tested, regardless of safety factor.
SMBA-256	Test levels for Beryllium structure shall be 1.25 times limit level for both qualification and acceptance testing.
SMBA-257	Test levels for composite structure, including metal matrix, shall be 1.25 times limit level for acceptance testing.
SMBA-258	For all elements that are to be qualified by analysis, positive strength margins on yield shall be shown to exist at stresses equal to 2.0 times those induced by the limit loads, and positive margins on ultimate shall be shown to exist at stresses equal to 2.6 times those induced by the limit loads.

7.4.2 Acoustic Testing

SMBA-259	An acoustic test shall be performed on instrument components unless an assessment of the hardware indicates that they are not susceptible to the expected acoustic environment, or responses are enveloped by random vibration testing, or that testing at higher levels of assembly provides sufficient exposure at an acceptable level of risk to the program.
SMBA-260	The instrument shall be designed to withstand the maximum expected launch vehicle flight acoustic environment during ascent as specified in Table 7.4.2-1 plus 3 dB.

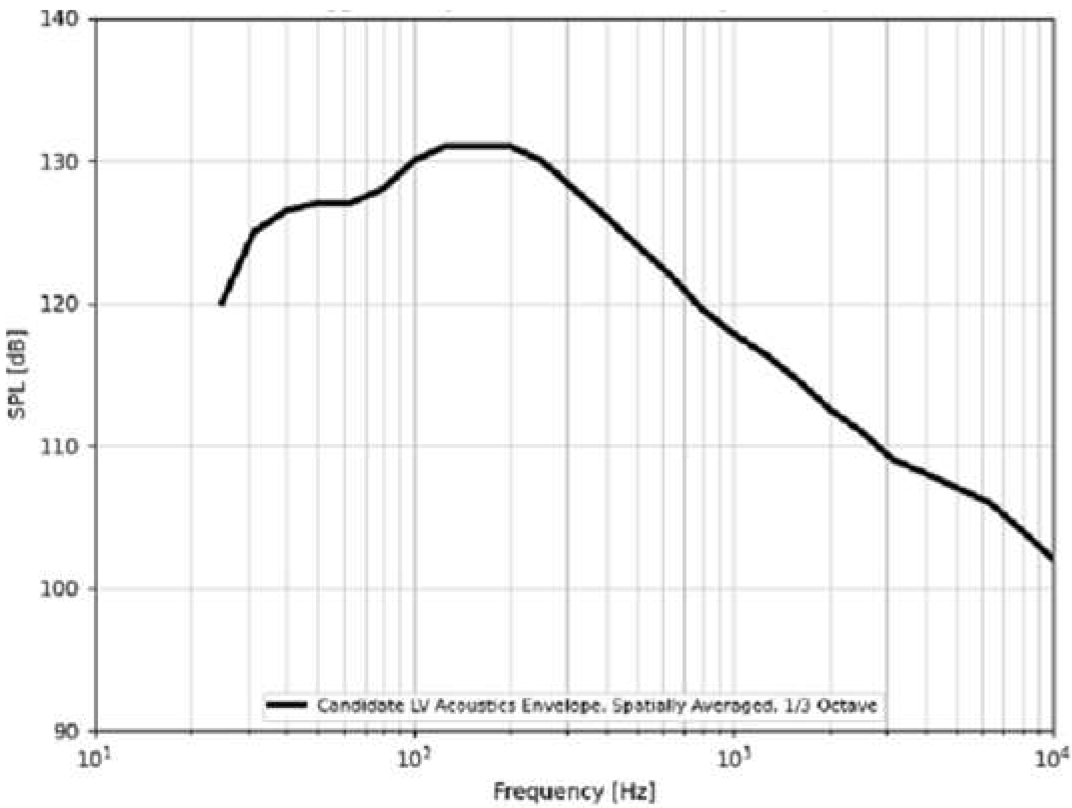

Figure 7.4.2-1. Payload Acoustics Maximum Predicted Environment (MPE) Design Envelope

Table 7.4.2-1. Payload Acoustics MPE Design Envelope

Candidate LV Payload Acoustics Design Envelope (P95/50 MPE)	
Frequency [Hz]	SPL [dB]
25	120
31.5	125
40	126.5
50	127
63	127
80	128
100	130
125	131
160	131
200	131
250	130
315	128
400	126
500	124
630	122
800	119.5
1000	117.8
1250	116.4
1600	114.5
2000	112.5
2500	111
3150	109
4000	108
5000	107
6300	106
8000	104
10000	102
OASPL[1]	140

Notes:
[1] Overall Sound Pressure Level

SMBA-261 Protoflight/qualification acoustic test levels shall increase the acceptance levels specified in Table 7.4.2-1 by 3 dB.

SMBA-262 The acoustic test duration shall be conducted for one minute (acceptance and protoflight) or two minutes (qualification).

7.4.3 Random Vibration Testing

SMBA-263 All instrument flight units shall be subjected to random vibration testing in three axes.

SMBA-264 The Instrument shall be designed to withstand the maximum expected launch vehicle flight random vibration environment during ascent as specified in Table 7.4.3-1 plus 3 dB.

Table 7.4.3-1. SMBA Acceptance Random Vibration Environment

SMBA Acceptance Random	
FREQ	**LEVEL**
20	0.010
50	0.03
950	0.03
2000	0.010
Overall	**6.74 Grms**

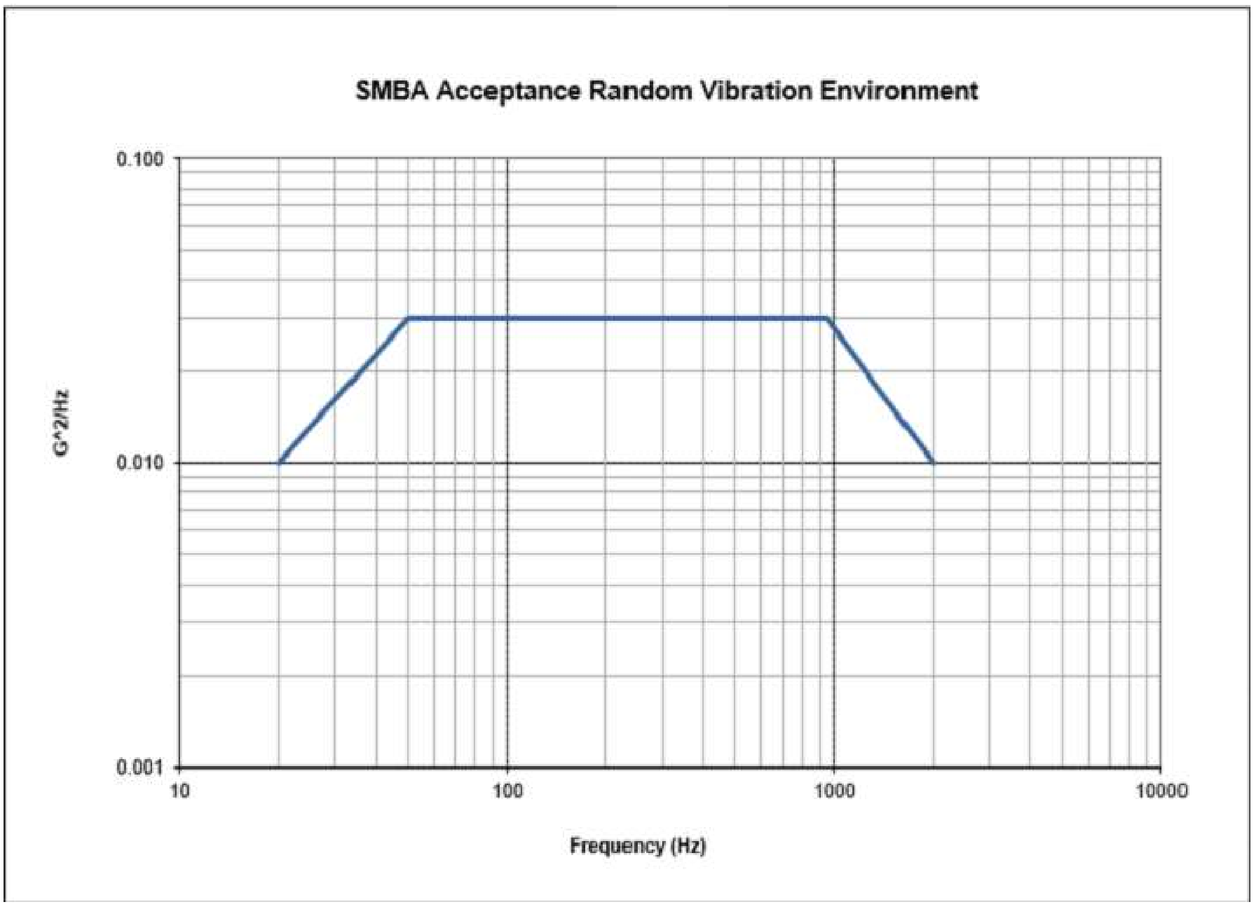

Figure 7.4.3-1. SMBA Acceptance Random Vibration Environment

SMBA-265 Random vibration testing shall be conducted for 1 minute per axis (Acceptance/Protoflight) and 2 minutes per axis (Qualification) with a spectrum that is based on expected levels at the component mounting locations.

SMBA-266 Protoflight and qualification random vibration test levels shall be 3 dB above the acceptance test levels.

SMBA-267 The acceptance test levels for any separately mounted component shall be greater than minimum workmanship levels specified in Table 7.4.3.1-1.

7.4.3.1 Random Vibration After Rework

SMBA-268 As a minimum, reworked components shall be subjected to a single axis workmanship random vibration test to the minimum workmanship test levels specified in Table 7.4.3-2 and Figure 7.4.3-2.

The determination of axis is to be made based on the direction necessary to provide the highest excitation of the reworked area. Testing may be required in more than one axis if a single axis test cannot be shown to adequately test all of the reworked area. If the amount of rework or disassembly required is significant, then 3-axis testing to acceptance levels may be necessary if they are higher than workmanship levels.

Table 7.4.3-2. Component Minimum Workmanship Random Vibration Test Levels

Frequency (Hz)	ASD Level (g^2/Hz)
20	0.01
20-80	+3 dB/oct
80-500	0.04
500-2000	-3 dB/oct
2000	0.01
Overall	6.8 grms

The plateau acceleration spectral density (ASD) level may be reduced for components weighing between 45.4 and 182 kg, or 100 and 400 pounds according to the component weight (W) up to a maximum of 6 dB as follows:

	Weight in kg	Weight in lb
dB reduction	=10 log (W/45.4)	=10 log (W/100)
ASD(plateau) level	=0.04•(45.4/W)	=0.04•(100/W)

SMBA-269 The sloped portions of the spectrum shall be maintained at plus and minus 3 dB/oct. Therefore, the lower and upper break points, or frequencies at the ends of the plateau become:
FL = 80 (45.4/W) [kg] FL = frequency break point low end of plateau
= 80 (100/W) [lb]
FH = 500 (W/45.4) [kg] FH = frequency break point high end of plateau
= 500 (W/100) [lb]

SMBA-270 The test spectrum shall not go below 0.01 g^2/Hz. For components whose weight is greater than 182 kg or 400 pounds, the workmanship test spectrum is 0.01 g^2/Hz from 20 to 2000 Hz with an overall level of 4.4 g_{rms}.

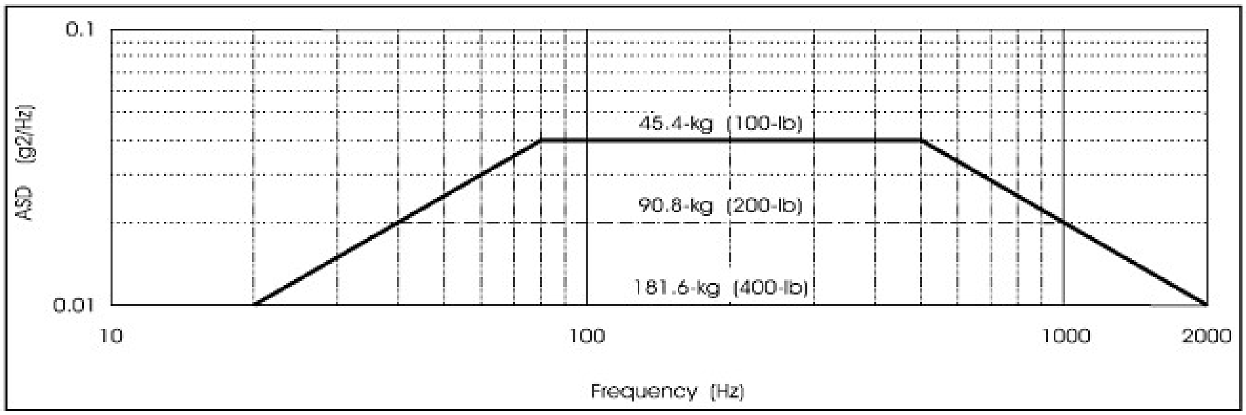

Figure 7.4.3-2. Minimum Workmanship Random Vibration Test Levels

7.4.4 Sine Vibration Testing

SMBA-271 All instrument flight units shall be subjected to sine sweep testing in three axes.

SMBA-272 For all component-level, instrument-level and spacecraft-level tests, the hardware under test shall be in the launch configuration. (If unit is powered during launch, the unit should be powered during testing.)

SMBA-273 There shall be one sweep from 5 Hz to 50 Hz for each axis.
Note: This is the typical frequency range of the sine test, but the range may be extended depending on the specific launch vehicle and the frequency content of the coupled loads analysis.

SMBA-274 The test sweep rate shall be 4 octaves/min for acceptance and protoflight and 2 octaves/min for qualification.

SMBA-275 The instrument shall be designed to withstand 1.25 times the maximum expected launch vehicle sine vibration environment values as specified in Table 7.4.4-1 in all three axes.

Table 7.4.4-1. SMBA Acceptance Sine Environment (All Axes)

Frequency (Hz)	Acceleration (g)
5 to 100	10

Notes:
[1] Input levels may be notched to limit subsystem/component CG response to the design limit loads.
[2] Low-frequency amplitude can be adjusted to stay within the stroke capability of the test facility.
[3] Hardware shall show positive margins for acceptance level Sine vibration using the factors of safety defined in Table 5.7.2-1.
[4] Acceleration levels applicable to hardware less than 200 kg.

SMBA-276	Protoflight/qualification sine vibration test levels shall be 1.25 times the acceptance test levels defined in Table 7.4.4-1.

SMBA-277	The instrument-level sine vibration tests shall be conducted to protoflight test levels unless demonstrated on an applicable qualification unit.

7.4.5 Shock Testing

Shock testing will be performed at the satellite level for each of the shock environments expected during launch/commissioning including payload attach fitting separation shocks, solar array deployment shocks, appendage/mechanism deployment shocks, and instrument-induced shocks (if required).

SMBA-278	The instrument shall be designed to withstand the maximum launch vehicle separation shock environment specified in the NEON Sounder Spacecraft to SMBA Instrument Interface Control Document ICD (NS1-SMBA-xxx-xxxx) on the instrument side of the mechanical mating interface with the satellite plus 3 dB. Note: 2 actuations are expected for qualification and protoflight and 1 for acceptance testing. Load limits are 1.4 times, 1.4 times, and 1 times the limit level respectively.

SMBA-279	A shock susceptibility assessment shall be performed on all instrument components. If the flight shock environment as shown on an SRS plot (Q=10) is enveloped by the curve shown in Figure 7.4.5-1, then the shock environment can be considered benign and there is low risk in deferring the shock test. For the case in which the shock levels are above the curve, then component level shock testing should be considered.

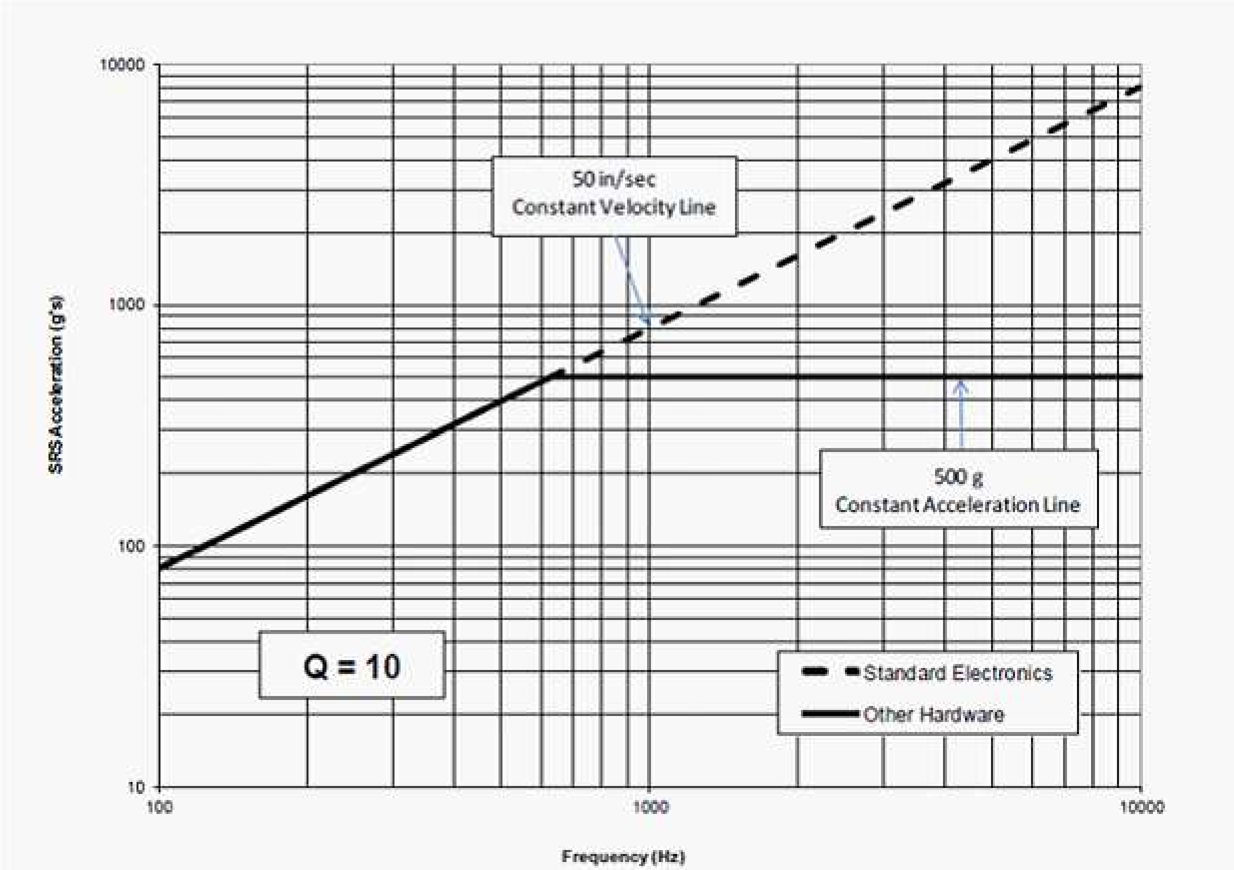

Figure 7.4.5-1. Shock Response Spectrum (SRS) for Assessing Component Test Requirements

SMBA-280	Instrument self-induced shock testing shall be accomplished by two actuations at the instrument level for each self-induced shock source (in order to account for the scatter associated with the actuation of the device) for the first flight unit, and a single actuation on subsequent units.

7.5 Mechanism and Deployment Verification

SMBA-281	For deployment circuits, the Instrument shall demonstrate that electrical current margins on non-explosive initiator circuits will be met.

SMBA-282	Mechanisms shall be life tested to at least two times the number of planned operational cycles plus two times the total number of ground cycles (including assembly, installation, and maintenance, functional, environmental, and run-in cycles) without any failure.

7.6 Thermal Vacuum Environmental Testing

7.6.1 Thermal Vacuum Cycling

SMBA-283	Thermal vacuum testing shall be performed to verify the thermal design margins to the operational temperature ranges. Note: Thermal design margins may be defined using GSFC-STD-7000 Section 2.6.3 as a guideline.
SMBA-284	All instrument hardware shall be subjected to a minimum equivalent of 12 thermal vacuum temperature cycles before launch. Following integration of the Government-furnished instruments, the integrated satellite will be subjected to a minimum of four cycles of thermal vacuum testing.
SMBA-285	To verify survival performance, thermal vacuum testing shall include a minimum of one plateau at each survival temperature extreme for a minimum of 8 hours. Following each survival plateau, the unit shall be returned to within its operating temperature limits to verify performance.
SMBA-286	Temperature dwell (plateau) periods at the instrument level of assembly shall be greater than or equal to 12 hours at each plateau.
SMBA-287	During the thermal vacuum cycling with exception to the survival plateaus, the hardware shall be operating, and its performance evaluated against specifications during transitions and temperature plateaus.
SMBA-288	All instrument systems that will be ON during launch shall be powered prior to starting chamber pumpdown and their performance verified during transition through the voltage breakdown/electrical discharge region.
SMBA-289	The chamber pressure after the electrical discharge checks are conducted shall be less than 1.33×10^{-3} Pa (1×10^{-5} torr).
SMBA-290	Thermal vacuum tests shall demonstrate satisfactory operation over the range of possible flight voltages.
SMBA-291	Thermal vacuum testing shall demonstrate satisfactory operation of the instrument in all modes including Safe and Survival and include demonstrations of Safe Mode recovery.
SMBA-292	Thermal vacuum testing shall demonstrate satisfactory operation of the instrument survival heaters, both primary and redundant, to extent possible while at survival temperature.
SMBA-293	Hot and cold turn-on shall be demonstrated during thermal vacuum tests.

SMBA-294	For actively controlled systems such as cold plates, heaters, thermoelectric coolers, loop heat pipes, capillary pumped loops, or other devices with selectable/variable set points, a test temperature margin of 5 °C shall be imposed on the respective interface temperature that is under control.

7.6.2 Thermal Design Margin (Thermal Balance) Testing

SMBA-295	Thermal balance testing shall be performed at worst-case hot operational, worst-case cold operational, cold survival, and nominal operational, using the environmental temperature ranges. Note: Thermal balance testing may be defined using GSFC-STD-7000 Section 2.6.4 as a guideline.

SMBA-296	Thermal design margins shall be verified under worst-case hot operational and cold operational, survival, and, if tested, Safe Mode conditions.

Stabilization may be considered to have been achieved when all temperature sensors indicate +/- 0.1 °C/hour for at least 4 consecutive hours.

SMBA-297	Acceptance ranges shall be calculated by adding a thermal margin of ±5 deg C to the maximum expected temperature range per Figure 7.6.2-1. Note: For subsystems and lower-level components, this results in acceptance test levels being ±10 deg C on top of the analytically determined extreme temperatures predicted from thermal models.

SMBA-298	Protoflight ranges shall be calculated by adding ±10 deg C to the maximum expected temperature range per Figure 7.6.2-1. Note: For subsystems and lower-level components, this results in protoflight test levels being ±15 deg C on top of the analytically determined extreme temperatures predicted from thermal models.

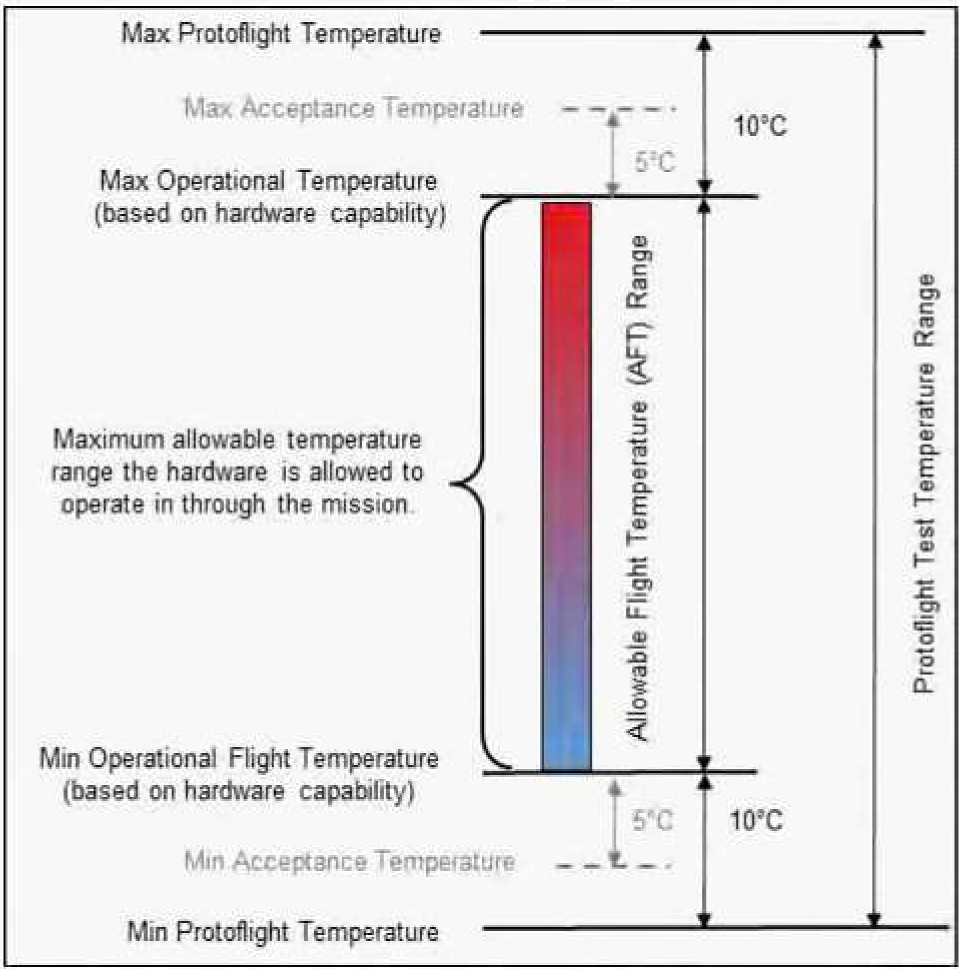

Figure 7.6.2-1. Qualification, Protoflight, and Flight Acceptance Thermal-Vacuum Temperatures

7.6.3 Contamination/Bake-Out Requirements

At the completion of instrument-level thermal vacuum testing, the Instrument should be thermally baked out following normal thermal vacuum multi-cycle functional testing at < 1E-5 torr during the last high temperature soak cycle followed by ambient temperature at test end.

Demonstration that the instrument has outgassed sufficiently should be given by using a minimum of seven Temperature Quartz Crystal Microbalances (either thermal or cryogenic) operating at a collection temperature of -20 °C to monitor out-gassing rate during bake out. All contamination bakeouts should include a qualitative molecular analysis by scavenger plate or coldfinger assembly.

SMBA-299	During the last hot cycle of the instrument-level thermal vacuum testing, the Instrument shall not outgas at a rate greater than 4.62E-7 gram/second (TBS). Outgassing rate should be maintained for at least 4 continuous hours with a resolution of better than 10 minutes.

Multilayer insulation blankets, harnesses, and cables, and GSE electronic boxes/modules to be included in satellite-level thermal vacuum testing should be baked out separately.

8 REQUIREMENTS VERIFICATION MATRIX

8.1 General Verification Requirements

Each verification will be carried out in accordance with approved procedures. See applicable contract requirements to determine which procedures require government approval.

In the case of verification by test, the procedure will identify the specific test instrumentation (hardware and/or software) and/or special test equipment used.

8.2 Verification Cross Reference

As summarized in Appendix A - SMBA PRD Verification Cross Reference Matrix (VCRM), each requirement of this specification will be verified by the method (or combination of methods) specified in the CDRL SE-4 Verification Matrix (NS1-SMBA-CDRL-0001, NEON Series-1 SMBA Instrument CDRL DID). The approved CDRL SE-4 Verification Matrix will become the initial Appendix A. Subsequent changes to Appendix A will occur through approved CDRL SE-4 updates as required and the GSFC Change Control Board (CCB) process.

8.3 Definition of Verification Methods

There are four acceptable requirement verification methods as defined below: Analysis, Inspection, Demonstration, and Test. Verification by Design is not an acceptable verification method. Generally, requirements that contained verification by design should be defined as Inspection of configuration-controlled design documentation, such as drawings or software.

8.3.1 Verification by Analysis

Verification by Analysis is the interpretation, interpolation, or extrapolation of analytical, empirical, or test data to show theoretical compliance with stated requirement. This method also applies to requirements to perform an analysis, or that specify how an analysis is to be performed.

8.3.2 Verification by Inspection

Verification by Inspection is an observation or examination of the item that does not require the item to be powered or operating. This includes:

a. Certain requirements verified by simple mechanical measurements that do not require the item to be powered or operated.
b. Hardware implementation constraints (such as required or prohibited materials or processes).
c. Software implementation constraints (such as coding standards) – verified by examination of source code.
d. Requirements for documentation and other requirements that are verified by the examination of documentation. The documentation should be configuration controlled at time of inspection.

8.3.3 Verification by Demonstration

Verification by Demonstration is an exhibition of the operability or supportability of an item under either controlled or intended service-use conditions. These verifications are oriented almost exclusively toward acquisition of qualitative data, not requiring use of special test equipment or processing of data outputs from the hardware.

8.3.4 Verification by Test

Verification by Test is an action that verifies an item's operability, supportability, performance capability, or other specified qualities when subjected to controlled conditions that are real or simulated. These verifications require use of special test equipment and sensors to obtain quantitative data for analysis as well as qualitative data derived from displays and indicators inherent in the item(s) for monitor and control. The Test method includes examination of output data from the unit under test that has been collected and/or processed by special test equipment. This also applies to requirements to perform a test, or that specify how a test is to be performed.

Appendix A Verification Cross Reference Matrix (VCRM)

TBS

Appendix B Acronyms/Abbreviations

°	Degrees
°C	Degrees Celsius/Centigrade
°F	Degrees Fahrenheit
$\Delta\Phi$	Change in angle
ΔT	Change in time
λ	Wavelength
σ	Sigma
μ	Micro
μF	Microfarad
$\mu g/cm^2$	Micrograms per square centimeter
μm	Micrometer
μs	Microseconds
a	Acceleration
A	Ampere
ADC	Analog-to-Digital Converter
AFT	Allowable Flight Temperature
AR	Attitude Reference
ASD	Acceleration Spectral Density
ASTM	American Society for Testing and Materials
ATMS	Advanced Technology Microwave Sounder
AVMP	Atmospheric Vertical Moisture Profile
AVTP	Atmospheric Vertical Temperature Profile
BW	Bandwidth
C	Celsius/Centigrade
C&DH	Command and Data Handling
C&T	Command and Telemetry
CCB	Change Control Board
CCSDS	Consultative Committee for Space Data Systems
CDRL	Contract Data Requirements List
CE	Conducted Emission
CG	Center of Gravity
cm	Centimeter
CM	Common Mode
CMCE	Common Mode Conducted Emissions
CMD	Command
CONOPS	Concept of Operations
CPT	Comprehensive Performance Test
CPU	Central Processing Unit
CrIS	Cryogenic Infrared Sensor
CS	Conducted Susceptibility
d	Distance

dB	Decibel
dBm	Decibels Referenced to 1 milliwatt
dBµA	Decibels Micro Amp
dBµV	Decibels Micro Volt
dBµV/m	Decibels Micro Volt per meter
DC	Direct Current
DID	Data Item Description
DM	Direct Mode
E-Field	Electric Field
ECEF	Earth Centered Earth Fixed
ECI	Earth Centered Inertial
EDR	Environmental Data Record
EEE	Electronic and Electrical Equipment
EEPROM	Electrically Erasable Read Only Memory
EGSE	Electronic Ground Support Equipment
ELV	Expendable Launch Vehicle
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
f0	Tuned Frequency
FET	Field Effect Transistor
FH	Frequency upper break point
FL	Frequency lower break point
FS	Factor of Safety
FTS	Flight Test System
g	gram
g	gravity
G^2/Hz or g^2/Hz	Acceleration spectral density
G	Giga
GCT	NEON Sounder Ground System Compatibility Test
GEVS	General Environmental Verification Specification
GHz	Gigahertz
GN&C	Guidance, Navigation, and Control
GPS	Global Positioning System
Grms	Root Mean Square Acceleration
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
Hz	Hertz
I	Moment of Inertia
I&T	Integration and Test
ICD	Interface Control Document
IEST	Institute of Environmental Sciences and Technology
ITU	International Telecommunication Union

JPSS	Joint Polar Satellite System
JSC	Johnson Space Flight Center
K	Kelvin
k	Kilo
kg	Kilogram
kHz	Kilohertz
km	Kilometer
krad	Kilorad
L2	L-Band 2 used in navigation communications
lb	Pound
LEO	Low Earth Orbit
LISN	Line Impedance Stabilization Network
LPT	Limited Performance Test
LTAN	Local Time Ascending Node
LV	Launch Vehicle
m	Meter
M	Mega
Mbps	Mega bits per second
MHz	Megahertz
MICD	Mechanical Interface Control Document
MIL-STD	Military Standard
mm	Millimeter
MPE	Maximum Predicted Environment
msec	millisecond
MUF	Model Uncertainty Factor
MWS	Microwave Sounder
N-m	Newton-meter
NASA	National Aeronautics and Space Administration
NEDT	Noise Equivalent Temperature Difference
NESDIS	National Environmental Satellite, Data, and Information Service (NOAA)
NEON	Near Earth Orbit Network
NIST	National Institute of Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NS1	NEON Series-1 Project
NSOSA	NOAA Satellite Observation System Architecture Study
NTE	Not to exceed
NWP	Numerical Weather Prediction
OASPL	Overall Sound Pressure Level
OBC	On-Board Computer
Oct	Octave
oz-in	Ounce - inch
Pa	Pascal

PAC	Percent Area Coverage
PLAR	Post-Launch Assessment Review
PPF	Payload Processing Facility
PROM	Programmable Read-Only Memory
PRD	Performance Requirements Document
Pwr	Power
q	Water content or humidity
Q	Quality
RAM	Random Access Memory
RDR	Raw Data Record
RE	Radiated Emissions
REC	Receive
RF	Radio Frequency
RFI	Radio Frequency Interference
RMS	Root Mean Square
RS	Radiated Susceptibility
RSS	Root Sum Squared
SC	Spacecraft
S/C	Spacecraft
SDE	Scan Drive Electronics
SDR	Sensor Data Record
sec	Second
Si	Silicon
SI	International System of Units (Metric)
SIIS	Spacecraft Instrument Interface Simulator
SMBA	Sounder for Microwave-Based Applications
SOW	Statement of Work
sqrt	Square root
SPL	Sound Pressure Level
SRF	Spectral Response Function
SRS	Shock Response Spectrum
SSO	Sun Synchronous Orbit
T	Temperature
T	Torque
T_{avail}	Minimum available torque under worst case conditions at any time of life
TBD	To Be Determined
TBR	To Be Reviewed
TBS	To Be Supplied
TDR	Temperature Data Record
TID	Total Ionizing Dose
TLM	Telemetry
TM	Torque Margin
TMON	Telemetry Monitoring

TOD	Time of Day
UTC	Coordinated Universal Time
V	Volt
VCRM	Verification Cross Reference Matrix
VDC	Volts Direct Current
W	Watt
W	Weight